



Report for IC-Technology Lab



Group 9

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1 Introduction

This report presents the results of the ET4ICP IC-Technology Practical Course and studies how the BICMOS5 fabrication process determines the final electrical behaviour of integrated-circuit test structures and MOS transistors. The work combines analytical calculations, process and device simulations, cleanroom processing assignments, and final electrical measurements on fabricated wafer structures.

The practical course is centered on the educational BICMOS5 process of the Else Kooi Laboratory, in which NMOS and PMOS devices are formed together with process control module (PCM) test structures for monitoring the technological steps. The process uses five masks (N-Well, Shallow-N, Shallow-P, contact openings and interconnect) plus an unmasked boron threshold-adjust implantation whose dose is varied over the four quarters of the wafer. Besides the transistors themselves, the wafer contains dedicated structures such as Van der Pauw resistors and electrical linewidth measurement structures, which make it possible to extract sheet resistance, doping-related quantities, and lateral diffusion effects from the finished process.

The main goal of this report is to document the assignments of the course and to compare results obtained from theory, Sentaurus simulations, processing observations, and electrical measurements. In particular, the report relates the analytical calculations and simulations of junction depth, implantation behaviour, and threshold-voltage adjustment to the experimental results for sheet resistance, dopant concentration, transistor threshold voltage, mobility, and short-channel effects.

The report is organized as follows. Section 2 presents the technology assignments and analytical calculations based on the BICMOS5 process flow. Section 3 discusses the process simulations, including implantation, diffusion, channeling, and masking behaviour. Section 4 covers the device simulations of the NMOS and PMOS structures, with emphasis on junction depths, threshold voltage, and current-voltage characteristics. Section 5 presents the cleanroom processing assignments, including doping concentration extraction, oxide deposition and etching, and aluminium patterning. Section 6 reports the electrical measurements on the fabricated BICMOS5 wafer, including Van der Pauw, ELM, and transistor characterization. Finally, Section 7 compares the overall trends and discusses the agreement and remaining differences between analytical models, simulations, processing results, and measurements.

2 Technology Assignments

This section covers the five preparatory assignments on the BICMOS5 process [1]. They establish the analytical baseline of the report: the alignment-marker step height from the Deal-Grove oxidation model, the choice of gate oxidation, the junction depths of the three implantations, the mask order of the process flow, and the theory of the V_T -adjust implantation. The junction depths calculated here are compared with the Sentaurus simulations in Section 4, and the V_T -adjust theory is verified by the measurements in Section 6.

Assignment 1: Marker oxidation

The waferstepper alignment markers are the first step of the process. For the waferstepper to see these markers with a laser, a step has to be formed in the silicon. A way of forming this step is by performing a double oxidation. Here the step height made in the Si wafer during this process is calculated.

The oxidation rate of silicon is given by the following equation (Deal-Grove model [2]):

$$\frac{dx}{dt} = \frac{B}{2x + A}$$

where x = oxide thickness, A and B are constants.

a) Show that the solution of this equation for t with $x = x_0$ at $t = 0$ as initial condition is:

$$t = \frac{(x^2 - x_0^2) + A(x - x_0)}{B}$$

Separating the variables of

$$\frac{dx}{dt} = \frac{B}{2x + A}$$

gives

$$(2x + A)dx = Bdt$$

Integrating the left side from x_0 to x and the right side from 0 to t yields

$$\int_{x_0}^x (2x + A)dx = \int_0^t Bdt$$

$$(x^2 - x_0^2) + A(x - x_0) = Bt$$

Therefore, the solution of this equation for t with $x = x_0$ at $t = 0$ as initial condition is

$$t = \frac{(x^2 - x_0^2) + A(x - x_0)}{B}$$

At 1100°C the constants are: $B = 0.553 \mu\text{m}^2/\text{hr}$ and $B/A = 3.1695 \mu\text{m}/\text{hr}$.

b) Calculate the oxide thickness after the first oxidation (time=38.5 min).

Now a window is etched in the SiO_2 to make the marker, as shown in Figure 1, after which another oxidation is performed.

From $B = 0.553 \mu\text{m}^2/\text{hr}$, $\frac{B}{A} = 3.1695 \mu\text{m}/\text{hr}$, so $A = \frac{B}{B/A} \approx 0.1745 \mu\text{m}$, and with $t = 38.5 \text{ min} \approx 0.6417 \text{ hr}$ and $x_0 = 0$ before the first oxidation,

$$t = \frac{x^2 + Ax}{B}$$

$$x^2 + Ax - Bt = 0$$

$$x^2 + 0.1745x - 0.3548 = 0$$

This equation gives two roots, which are $x \approx 0.5147 \mu\text{m}$ and $x \approx -0.6893 \mu\text{m}$ (discarded). Therefore, the first oxide thickness is $0.5147 \mu\text{m}$.

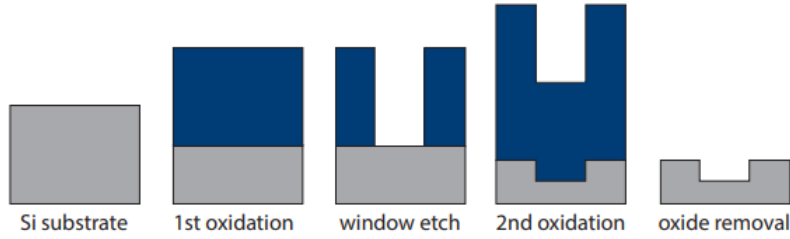


Figure 1: The double-oxidation sequence used to form the alignment marker: first oxidation, window etch, second oxidation, and full oxide removal.

c) Calculate the oxide thickness after the second oxidation (time=38.5 min) for both the etched and non-etched areas.

For the second oxidation, the initial oxide starts from $x_0 = 0$ for the etched area, so the second oxide thickness for the etched area at 38.5 min is again $0.5147 \mu\text{m}$.

For the non-etched area, the initial oxide starts from $x_0 = 0.5147 \mu\text{m}$. From

$$t = \frac{(x^2 - x_0^2) + A(x - x_0)}{B}$$

$$x^2 + Ax - x_0^2 - Ax_0 - Bt = 0$$

$$x^2 + 0.1745x - 0.2649 - 0.0898 - 0.3548 = 0$$

$$x^2 + 0.1745x - 0.7095 = 0$$

Solving the equation above gives two roots, which are $x \approx 0.7596 \mu\text{m}$ and $x = -0.9341 \mu\text{m}$ (discarded). So after another 38.5 min, the oxide thickness for the non-etched area is $0.7596 \mu\text{m}$.

During oxidation silicon is converted into silicon dioxide. Hence the silicon wafer gets thinner. For a certain thickness of oxide d_{ox} the amount of silicon consumed is $0.455 d_{ox}$.

d) After all the oxide is removed there is a step in the silicon. Calculate the height h of this step (see Figure 1).

For an oxide thickness d_{ox} , a silicon thickness of $d_{si} = 0.455 d_{ox}$ is consumed during the oxidation. In the marker window the silicon is consumed twice, once per oxidation, for a total grown oxide of $0.5147 + 0.5147 = 1.0294 \mu\text{m}$, while in the non-etched area silicon is only consumed up to the final oxide thickness of $0.7596 \mu\text{m}$. After stripping all the oxide, the window therefore sits lower than its surroundings by

$$h = 0.455 (1.0294 - 0.7596) \approx 0.123 \mu\text{m}, \quad (1)$$

so the alignment marker is a recess of about 123 nm in the silicon surface.

Assignment 2: Gate oxidation

As explained above the gate oxide is formed during the anneal which activates the implanted SN and SP areas. It is important that this oxide is of good quality and the thickness is accurate. Moreover, during this oxidation the implanted atoms diffuse, which could cause undesired effects. Explain what kind of oxidation you would choose to perform this step in the BICMOS5 process (wet or dry oxidation).

Dry oxidation should be used for the gate oxide, for two reasons. First, quality: dry oxide is dense and has a low defect density and a clean Si/SiO₂ interface, whereas wet oxide incorporates OH groups that act as interface traps and oxide charge, which would shift the threshold voltage and degrade the channel mobility, exactly the parameters the gate oxide must keep well defined. Second, control: wet oxidation is roughly an order of magnitude faster, because water has a higher solubility than O₂ in SiO₂, so the slow dry growth lands much more reliably on the 100 nm target thickness. Dopant diffusion of the implanted SN and SP regions occurs in both cases, but the anneal is short enough that the profiles do not smear out significantly.

Assignment 3: Implantation and diffusion

In total three implantations are used in the BICMOS5 process to form the different n and p-type areas. At a place where the concentration of n-type dopants (N_D) equals that of p-type dopants (N_A) they cancel each other out. At these locations the p-n junctions will be located in the Si, as illustrated in Figure 2.

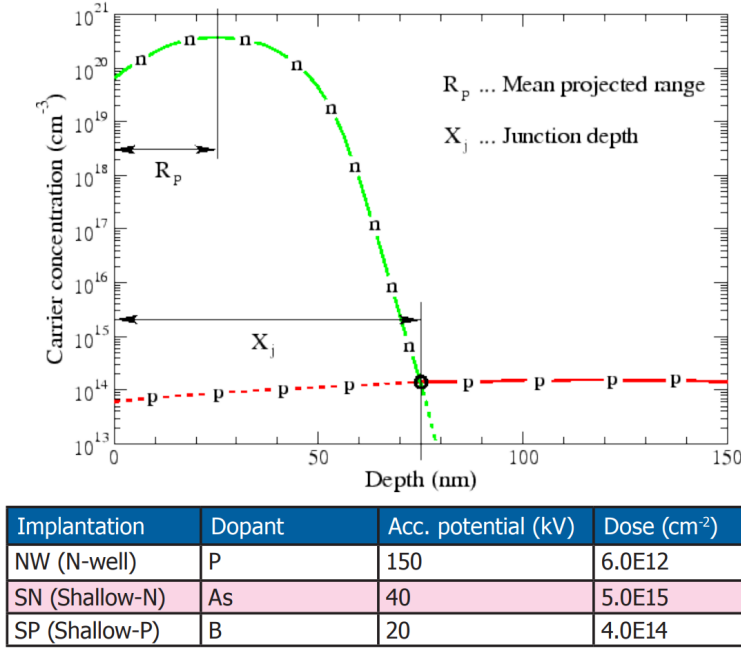


Figure 2: Definition of the projected range R_p and junction depth x_j : the junction lies where the implanted profile crosses the background doping [1].

Knowing the depths of these junctions is required in order to evaluate the electrical measurements of Sections 5 and 6. Here the junction depths are approximated with simple calculations, which are later compared with the simulations of Section 4. The implantation parameters are: N-Well (P, 150 kV, $6.0 \times 10^{12} \text{ cm}^{-2}$), SN (As, 40 kV, $5.0 \times 10^{15} \text{ cm}^{-2}$) and SP (B, 20 kV, $4.0 \times 10^{14} \text{ cm}^{-2}$).

a) The first implantation is the N-Well. This implantation is also followed by a 4 hours anneal at 1150°C to drive the P atoms deeper into the surface. If we ignore the implantation depth, we can approximate this process as a limited source diffusion. At 1150°C the diffusion constant of phosphorus is $8.86 \times 10^{-13} \text{ cm}^2/\text{s}$. Calculate the peak concentration of the implantation and then the junction depth after diffusion, assuming the silicon wafer has a p-type concentration of $10^{16} \text{ atoms/cm}^3$.

If the implantation depth is ignored, the N-Well implantation followed by its anneal can be approximated as a limited source diffusion. The approximation is justified because the implanter delivers a fixed dose Q and no source of new dopant atoms is present during the anneal (unlike a doped-oxide constant-source diffusion), so a fixed amount of dopant simply redistributes.

For limited source diffusion, the solution of the diffusion equation is a Gaussian profile,

$$C(x, t) = \frac{Q}{\sqrt{\pi Dt}} e^{-\frac{x^2}{4Dt}}, \quad (2)$$

where C is the dopant concentration at depth x after diffusion time t , Q the implanted dose, and D the diffusion constant.

From the given data, $Q = 6.0 \times 10^{12} \text{ cm}^{-2}$, $D = 8.86 \times 10^{-13} \text{ cm}^2/\text{s}$ and $t = 4 \text{ hr} = 14400 \text{ s}$. The peak concentration occurs at $x = 0$, so

$$C_{max} = \frac{Q}{\sqrt{\pi Dt}}$$

$$C_{max} = \frac{6.0 \times 10^{12}}{\sqrt{\pi(8.86 \times 10^{-13})(14400)}} \approx 2.997 \times 10^{16} \text{ cm}^{-3}$$

With the substrate p-type concentration of 10^{16} atoms/cm³ given in the problem statement, the junction depth is where the phosphorus concentration equals the background p-type concentration:

$$C(x, t) = 1.0 \times 10^{16} \text{ cm}^{-3}$$

$$1.0 \times 10^{16} = 2.997 \times 10^{16} e^{-\frac{x^2}{4Dt}}$$

Rearrange the equation:

$$\frac{1.0 \times 10^{16}}{2.997 \times 10^{16}} = e^{-\frac{x^2}{4Dt}}$$

$$\ln \left(\frac{1.0 \times 10^{16}}{2.997 \times 10^{16}} \right) = -\frac{x^2}{4Dt}$$

$$x = \sqrt{-4Dt \ln \left(\frac{1.0 \times 10^{16}}{2.997 \times 10^{16}} \right)}$$

Substitute the values:

$$x = \sqrt{-4(8.86 \times 10^{-13})(14400) \ln \left(\frac{1.0 \times 10^{16}}{2.997 \times 10^{16}} \right)} \approx 2.367 \times 10^{-4} \text{ cm} = 2.367 \text{ } \mu\text{m}$$

Therefore, the N-Well junction depth is approximately 2.37 μm .

b) The second implantation is the SN and is performed using As in the p-type Si substrate with a concentration of 10^{16} boron atoms/cm³. Assuming a Gaussian implantation profile, a projected range (R_p) of 0.0314 μm , and straggle (ΔR_p) of 0.0114 μm , calculate the junction depth of the SN implantation using the Gaussian implantation distribution.

Assuming a Gaussian implantation profile, the distribution of the dopant atoms in the silicon is approximated by

$$N(x) = \frac{Q}{\sqrt{2\pi}\Delta R_p} e^{-\frac{(x-R_p)^2}{2\Delta R_p^2}}, \quad (3)$$

where N is the dopant concentration at depth x , Q the implanted dose, R_p the projected range and ΔR_p the straggle.

From the given data, $Q = 5.0 \times 10^{15} \text{ cm}^{-2}$, $R_p = 0.0314 \text{ } \mu\text{m}$ and $\Delta R_p = 0.0114 \text{ } \mu\text{m} = 1.14 \times 10^{-6} \text{ cm}$, the peak concentration is

$$N_{max} = \frac{Q}{\sqrt{2\pi}\Delta R_p} = \frac{5.0 \times 10^{15}}{\sqrt{2\pi}(1.14 \times 10^{-6})} \approx 1.75 \times 10^{21} \text{ cm}^{-3}$$

This peak is above the solid solubility of arsenic at the anneal temperature, so in reality part of the dose will not be electrically active; this is one of the simplifications of the analytical model.

The junction depth is where the arsenic concentration equals the background boron concentration:

$$N(x) = 1.0 \times 10^{16} \text{ cm}^{-3}$$

Therefore,

$$1.0 \times 10^{16} = 1.7497 \times 10^{21} e^{-\frac{(x-0.0314)^2}{2(0.0114)^2}}$$

$$\frac{1.0 \times 10^{16}}{1.7497 \times 10^{21}} = e^{-\frac{(x-0.0314)^2}{2(0.0114)^2}}$$

$$\ln \left(\frac{1.0 \times 10^{16}}{1.7497 \times 10^{21}} \right) = -\frac{(x-0.0314)^2}{2(0.0114)^2}$$

$$(x-0.0314)^2 = -2(0.0114)^2 \ln \left(\frac{1.0 \times 10^{16}}{1.7497 \times 10^{21}} \right)$$

$$x^2 - 2(0.0314)x + (0.0314)^2 - 0.0031379 = 0$$

$$x^2 - 0.0628x - 0.0021519 = 0$$

This equation gives two roots, which are $x \approx 0.0874 \mu\text{m}$ and $x \approx -0.0246 \mu\text{m}$ (discarded). Therefore, the SN junction depth is $0.0874 \mu\text{m}$.

c) Finally, the SP implantation is done using B. This implantation is performed inside the N-Well, of which you can assume that around the junction the doping concentration is 2×10^{16} phosphorus atoms/cm³. Given that $R_p = 0.0685 \mu\text{m}$ and $\Delta R_p = 0.0296 \mu\text{m}$, calculate the depth of this junction.

For the SP implantation, the Gaussian implantation profile of Equation (3) is used again.

From the given data, $Q = 4.0 \times 10^{14} \text{ cm}^{-2}$, $R_p = 0.0685 \mu\text{m}$ and $\Delta R_p = 0.0296 \mu\text{m} = 2.96 \times 10^{-6} \text{ cm}$, the peak concentration is

$$N_{max} = \frac{Q}{\sqrt{2\pi}\Delta R_p} = \frac{4.0 \times 10^{14}}{\sqrt{2\pi}(2.96 \times 10^{-6})} \approx 5.39 \times 10^{19} \text{ cm}^{-3}$$

The junction depth is where the boron concentration equals the local phosphorus concentration in the N-Well:

$$N(x) = 2.0 \times 10^{16} \text{ cm}^{-3}$$

Thus,

$$\begin{aligned} 2.0 \times 10^{16} &= 5.39 \times 10^{19} e^{-\frac{(x-R_p)^2}{2\Delta R_p^2}} \\ \ln\left(\frac{2.0 \times 10^{16}}{5.39 \times 10^{19}}\right) &= -\frac{(x-R_p)^2}{2\Delta R_p^2} \\ x &= R_p + \sqrt{-2\Delta R_p^2 \ln\left(\frac{2.0 \times 10^{16}}{5.39 \times 10^{19}}\right)} \end{aligned}$$

Substituting the values gives

$$x = 0.0685 + 0.0296 \sqrt{2 \ln\left(\frac{5.39 \times 10^{19}}{2.0 \times 10^{16}}\right)} = 0.0685 + 0.118 \approx 0.186 \mu\text{m}$$

Therefore, the SP junction depth is approximately $0.186 \mu\text{m}$. Table 1 summarizes the three analytical junction depths; they are compared with the process simulations in Section 4.

Table 1: Summary of the analytically calculated peak concentrations and junction depths.

Implantation	Dopant	$N_{max} \text{ (cm}^{-3}\text{)}$	$x_j \text{ (}\mu\text{m)}$
N-Well	P	3.0×10^{16}	2.37
SN	As	1.75×10^{21}	0.087
SP	B	5.39×10^{19}	0.186

Assignment 4: BICMOS process flow

Five parts of the masks from the BICMOS process are shown in Figure 3. Put the masks in the right processing order and indicate which number corresponds to which process step (e.g. mask 4 is the SP mask).

The five core masks of the BICMOS5 process are applied in the order NW $\rightarrow$ SN $\rightarrow$ SP $\rightarrow$ CO $\rightarrow$ IC: first the deep N-Well is formed, then the shallow n⁺ and p⁺ source/drain implantations, then the contact openings are etched in the gate oxide, and finally the aluminium interconnect is patterned. Matching the five PMOS top views of Figure 3 to this order: mask 3 is the NW mask (the large filled well area), mask 5 is the SN mask (the n⁺ guard-ring/well-contact outline), mask 4 is the SP mask (the p⁺ source/drain stripes inside the well), mask 2 is the CO mask (the small contact openings), and mask 1 is the IC mask (the metal

wiring with the SU, G, S and D pads). The processing order of the figures is therefore 3 (NW) $\rightarrow$ 5 (SN) $\rightarrow$ 4 (SP) $\rightarrow$ 2 (CO) $\rightarrow$ 1 (IC).

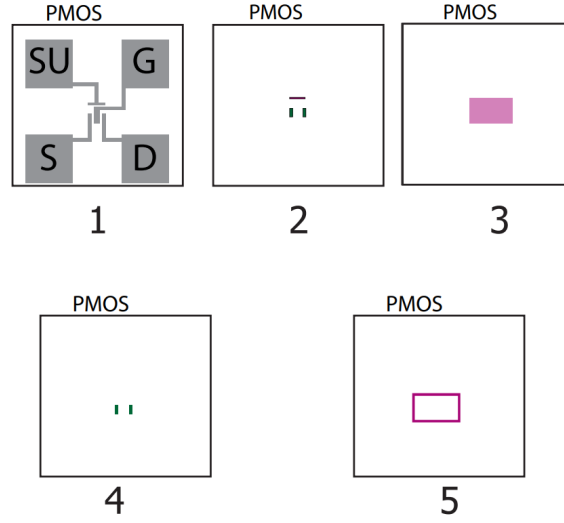


Figure 3: The five mask parts of the BICMOS5 process for a PMOS device, to be placed in processing order [1].

Assignment 5: V_T -adjust

Find the equations which describe the relation between the V_T and the doping levels in the channel for a PMOS and NMOS in literature, and explain why a boron implantation over the whole wafer works as a V_T adjustment for both devices. What happens to the threshold voltage of the NMOS when the V_T -adjust dose is increased? And for the PMOS?

For an NMOS with a channel acceptor concentration N_A , the relation between V_T and the doping level in the channel is given by [3]

$$V_T = V_{FB} + 2\phi_F + \frac{1}{C_{ox}} \sqrt{2\varepsilon_s q N_A \cdot 2\phi_F} \quad (4)$$

For a PMOS with a donor concentration N_D , the relation is

$$V_T = V_{FB} - 2|\phi_F| - \frac{1}{C_{ox}} \sqrt{2\varepsilon_s q N_D \cdot 2|\phi_F|} \quad (5)$$

In Equations (4) and (5), V_{FB} is the flat-band voltage, $\phi_F = (kT/q) \ln(N/n_i)$ the Fermi potential of the channel with doping N and intrinsic concentration n_i , C_{ox} the gate-oxide capacitance per unit area, ε_s the permittivity of silicon and q the elementary charge. The square-root term is the depletion charge that the gate must image before inversion.

Boron is a p-type dopant, and because the V_T -adjust implantation is performed without a mask, it adds a shallow sheet of acceptors near the surface in every channel on the wafer. In the NMOS channel the boron adds to the existing acceptor concentration, so N_A increases and V_T becomes more positive: the NMOS becomes harder to turn on. In the PMOS channel the boron partially compensates the donors, so the effective N_D decreases and V_T shifts towards zero: the PMOS becomes easier to turn on. A single boron implantation therefore increases the threshold magnitude of the NMOS and decreases that of the PMOS at the same time, which is exactly what is needed to give both devices threshold voltages of similar magnitude. Increasing the dose strengthens both effects, which is verified by the simulations in Section 4 and the measurements in Section 6.

3 Process Simulations

The doping profiles of the BICMOS5 process were simulated with Sentaurus Process (`sprocess`) on a one-dimensional grid: at every node the simulator tracks the dopant concentration through a sequence of implantation, diffusion, oxidation, deposition and etch statements. The assignments below investigate implantation profiles and channeling in crystalline and amorphous silicon, diffusion of the four standard dopants, the solid solubility limit, and the masking capability of photoresist, oxide and nitride layers. Assignments 1 and 2 of the manual concern inspecting and running the provided `implant-As.cmd`, `implant-B.cmd` and `implant-Sb.cmd` scripts; the numbering below follows the manual, starting with the first assignment that produces results for discussion. All implants use a dose of $1 \times 10^{15} \text{ cm}^{-2}$, with the energies chosen such that the projected ranges match. Each script implants a single species, because the lattice damage caused by one implant would change the channeling behaviour of the next one.

3.1 Process Simulation - 3

From the implantation profile plot in Fig. 4 and the generated `.out` files, the peak concentration for Sb is approximately $1.33 \times 10^{20} \text{ cm}^{-3}$ at $0.086 \mu\text{m}$ depth, for As it is about $1.13 \times 10^{20} \text{ cm}^{-3}$ at $0.091 \mu\text{m}$, and for B it is roughly $1.00 \times 10^{20} \text{ cm}^{-3}$ at $0.093 \mu\text{m}$.

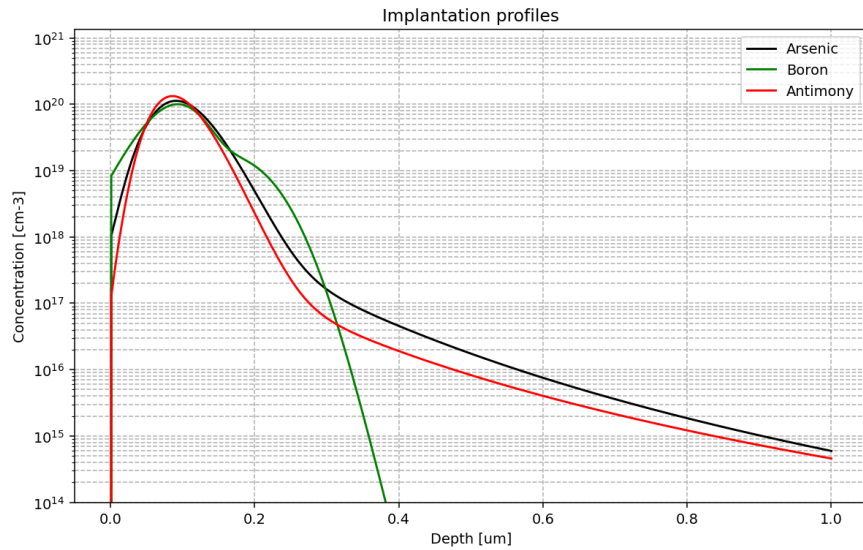


Figure 4: Simulated implantation profiles of arsenic (black), boron (green) and antimony (red) in crystalline silicon at matched projected range.

None of the profiles is Gaussian: a standard Gaussian would appear as a symmetric, inverted parabola on a log plot, whereas all three profiles show an asymmetric channeling tail towards larger depths. Among boron, arsenic and antimony, boron has the shortest tail, as its profile drops below 10^{14} cm^{-3} already around $0.4 \mu\text{m}$. Antimony has a longer tail than boron, but its concentration falls below that of arsenic at larger depths. Arsenic has the longest tail: its concentration is still the highest of the three at a depth of $1 \mu\text{m}$, a consequence of the ion-steering effect that guides As ions into channeling directions even for a tilted beam.

3.2 Process Simulation - 4

From `implant-B.cmd`, `implant-As.cmd` and `implant-Sb.cmd`, the implantation energies of B, As and Sb are 24 keV, 148 keV and 200 keV respectively. Since the atomic mass of P lies between those of B and As, one way to estimate the energy needed to implant P to the same projected range is linear interpolation between B and As:

$$E_P = E_B + \frac{m_P - m_B}{m_{As} - m_B}(E_{As} - E_B)$$

With the relative atomic masses of P, B and As being 31, 11 and 75 respectively, substitution gives

$$E_P = 24 + \frac{31 - 11}{75 - 11}(148 - 24) = 24 + \frac{20}{64} \times 124 \approx 62.75 \text{ keV}$$

Therefore, the energy used in `implant-P.cmd` is selected as $E_P \approx 63 \text{ keV}$.

Setting `energy=63` in `implant-P.cmd` and rerunning `implant.py` gives the plot shown in Fig. 5. The implantation profile of P is the blue curve.

From the generated .out file, the peak concentration of P, which is $1.19 \times 10^{20} \text{ cm}^{-3}$, appears at a depth of $0.080 \text{ }\mu\text{m}$. The resulting implantation profile has its peak at approximately the same depth as the other three dopants (Sb at $0.086 \text{ }\mu\text{m}$, As at $0.091 \text{ }\mu\text{m}$, B at $0.093 \text{ }\mu\text{m}$), so this energy is a reasonable estimate.

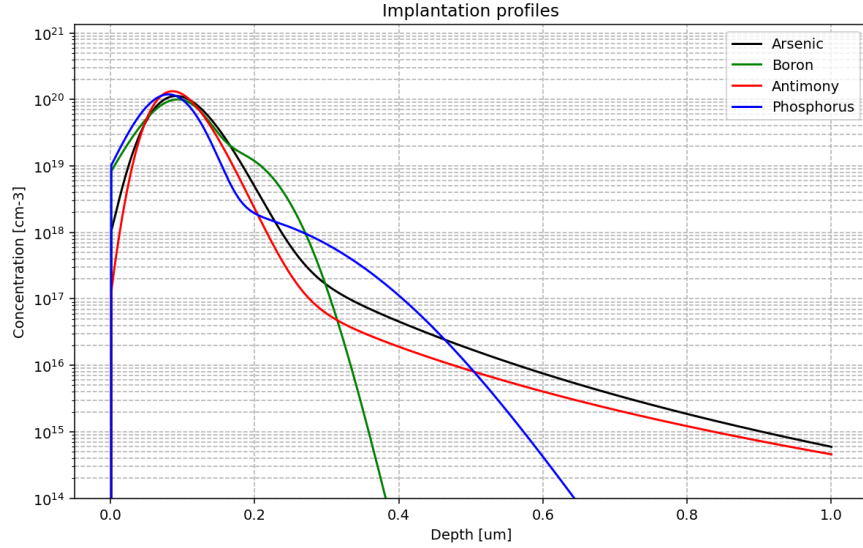


Figure 5: Simulated implantation profiles of arsenic (black), boron (green), phosphorus (blue) and antimony (red) in crystalline silicon at matched projected range.

3.3 Process Simulation - 5

The Monte Carlo simulated implantation profiles of boron, phosphorus, arsenic and antimony in amorphous silicon are shown in Fig. 6.

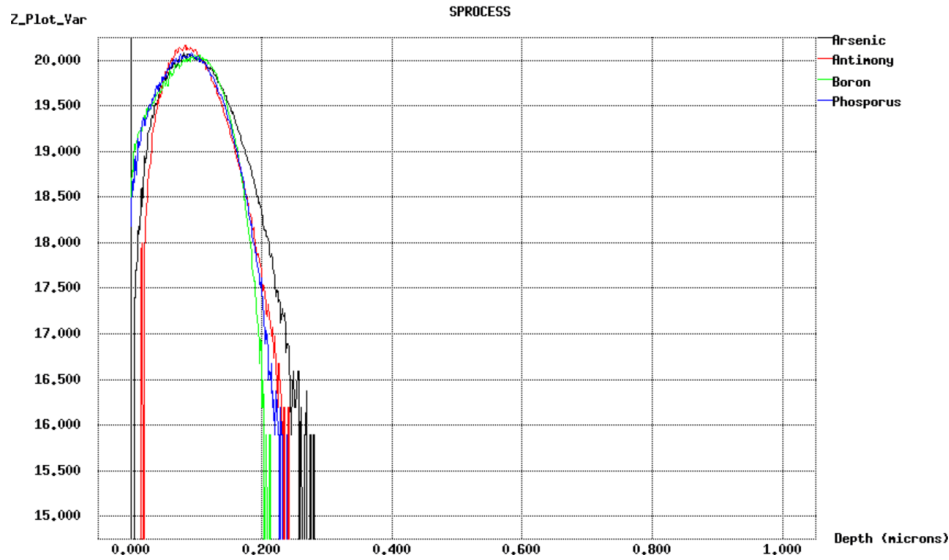


Figure 6: Monte Carlo simulated implantation profiles of the four dopants in amorphous silicon.

In amorphous silicon, all implantation profiles are closer to a Gaussian distribution. This is because amorphous silicon has no crystal channels, so channeling is strongly reduced. The implantation ions cannot travel deeply along specific lattice directions.

All implantations can be run after each other in amorphous silicon because the substrate has no crystal structure to begin with. The damage caused by one implantation does not change the channeling behaviour of the following implantation.

In contrast, the implantations in crystalline silicon were simulated separately because the damage from one implantation could affect the next implantation.

3.4 Process Simulation - 6 & 7

In this assignment, phosphorus is implanted at 40 keV while the tilt angle of the implantation is varied. Running the script with the default tilt of 7° gives the profile shown in Fig. 7.



Figure 7: Phosphorus implantation profile at 40 keV for the default tilt angle of 7° .

Using the generated .out files, the implantation profiles of phosphorus at 40 keV for six different tilt angles (0° , 3° , 7° , 11° , 13° and 15°) are compared in Fig. 8.

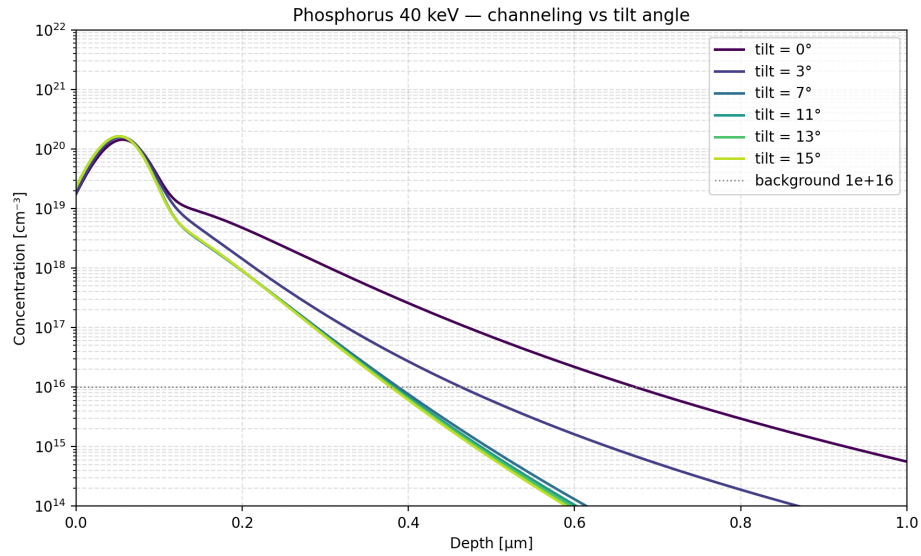


Figure 8: Phosphorus implantation profiles at 40 keV for tilt angles of 0°, 3°, 7°, 11°, 13° and 15°.

The tilt has a strong impact on the implantation depth: going from 0° to 7° nearly halves the depth at which the profile crosses the 10^{16} cm^{-3} level (from about 0.67 to 0.39 μm), because the tilted beam no longer looks straight into the open channels of the crystal. Beyond 7° the profiles hardly change, since the channels are already missed. The best angle found is therefore 7°: it is the smallest tilt that suppresses the channeling tail, and larger angles give no further improvement while increasing the lateral spread of the implant. This matches the standard 7° tilt and 22° rotation setting of the EKL implanter.

3.5 Process Simulation - 8

The dopant profiles after implantation in amorphous silicon followed by diffusion at 1000°C for 5 hours are shown in Figure 9.

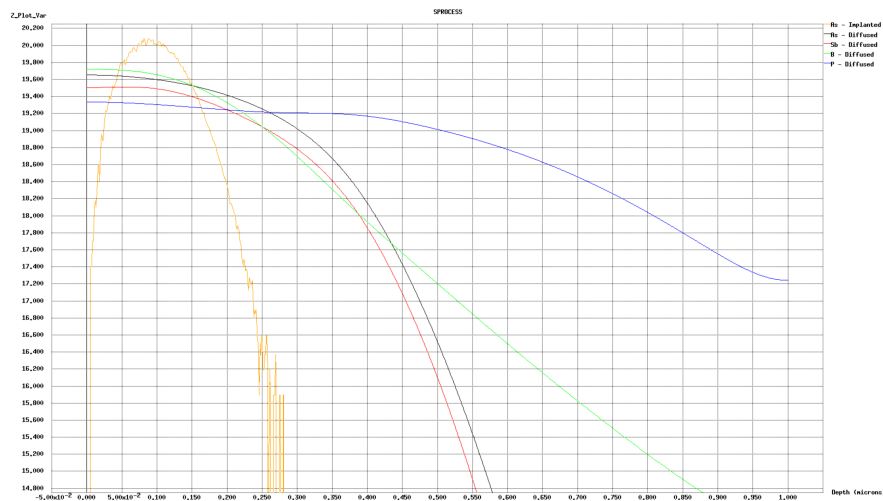


Figure 9: Diffused dopant profiles after 5 hours at 1000°C. The orange curve shows the implantation profile of As before diffusion.

Based on the simulated results, the dopants can be ordered from the narrowest to the widest diffusion profile as

$$\text{Sb} < \text{As} < \text{B} < \text{P}$$

Antimony and arsenic are relatively narrow, while boron and especially phosphorus diffuse much further into the silicon.

According to Table 2, dopants with negative misfit factors, e.g. boron and phosphorus, show stronger diffusion, while dopants with no misfit (As) or with positive misfit (Sb) diffuse less.

Table 2: Solid solubility limit at 1150 °C and misfit factor for different dopants [1].

Dopant	Solid solubility limit (cm^{-3})	Max active dopants (cm^{-3})	Radius ($\text{\AA}$)	Misfit factor
B	4×10^{20}	5×10^{19}	0.88	-0.254
P	1.5×10^{21}	1×10^{21}	1.10	-0.068
As	2×10^{21}	2×10^{21}	1.18	0
Sb	1×10^{20}	2×10^{19}	1.36	+0.153

Therefore, the simulation shows a relation between the misfit factor and the diffusion width, although the relation is clearly not proportional. This is because the misfit factor only affects the strain-driven component of the diffusion, while the dominant contribution comes from the diffusion mechanism itself: boron and phosphorus diffuse mainly via the fast interstitialcy mechanism, whereas arsenic and antimony are mostly substitutional diffusers that depend on the rarer vacancies [4].

3.6 Process Simulation - 9

The implantation and diffusion profiles of Sb for two different implantation doses ($1 \times 10^{15} \text{ cm}^{-2}$ and $1 \times 10^{14} \text{ cm}^{-2}$) are plotted.

For the default dose (Fig. 10), the implanted peak reaches $1.33 \times 10^{20} \text{ cm}^{-3}$, which is above the Sb solid solubility limit ($1 \times 10^{20} \text{ cm}^{-3}$, Table 2). After diffusion, the profile near the peak takes a flat-topped shape. The excess Sb above the limit cannot occupy lattice sites and forms inactive clusters, so the active concentration is held at the cap until the profile falls below the solubility limit.

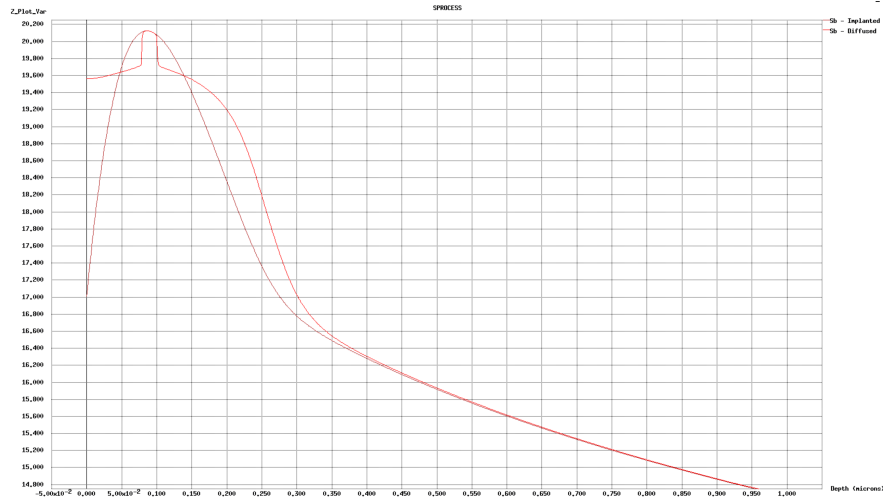


Figure 10: Antimony implantation and diffusion profile for a dose of $1 \times 10^{15} \text{ cm}^{-2}$.

For the reduced dose (Fig. 11), the peak of the implantation profile is $7.80 \times 10^{18} \text{ cm}^{-3}$, which is below the solubility limit, and the diffused profile therefore keeps a smooth, Gaussian-like shape without clipping.

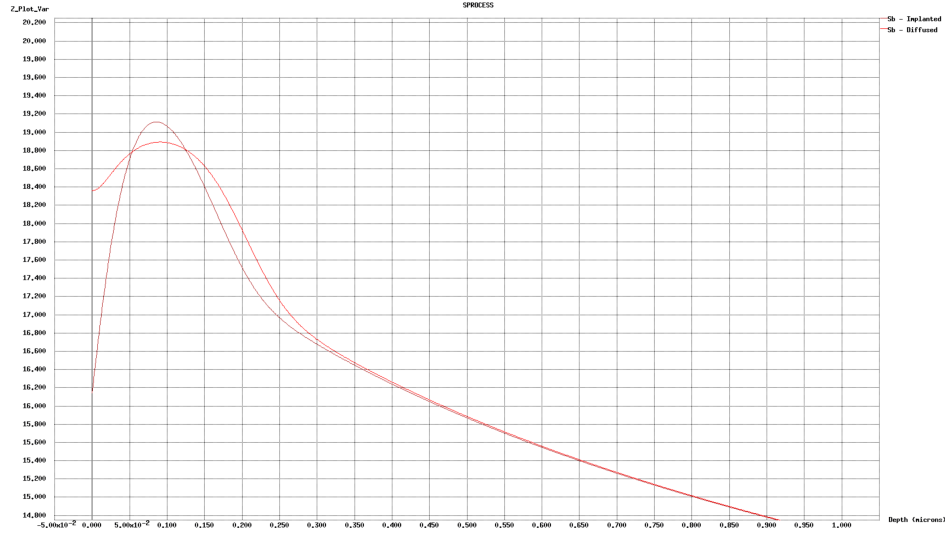


Figure 11: Antimony implantation and diffusion profile for a dose of $1 \times 10^{14} \text{ cm}^{-2}$.

3.7 Process Simulation - 10

Table 3 ranks the four dopants for each property from best (1) to worst (4), based on the simulation results of this section.

Table 3: Dopant ranking per property (1 = best, 4 = worst).

Dopant	Doping type	Short channeling tail	Narrow diffusion	Solid solubility
Sb	n	2	1	4
As	n	4	2	1
P	n	3	4	2
B	p	1	3	3

The channeling-tail ranking follows from the crystalline implantation profiles of Process Simulations 3 and 4, where the tail lengths (distance between the profile peak and the 10^{16} cm^{-3} crossing) order as $B < Sb < P < As$. The diffusion ranking follows from the diffused profile widths of Process Simulation 8 ($Sb < As < B < P$), and the solid solubility ranking from Table 2 ($As \times 10^{21}$, $P \ 1.5 \times 10^{21}$, $B \ 4 \times 10^{20}$, $Sb \ 1 \times 10^{20} \text{ cm}^{-3}$).

3.8 Process Simulation - 11

For the NMOS drain, a high dopant concentration and a sharp profile are required. Therefore, arsenic is the best choice. Arsenic is an n-type dopant with high solid solubility, which allows a high active concentration. It also diffuses narrowly, which gives a highly doped and relatively sharp source/drain junction.

For the N-Well, where the dopant must diffuse a few micrometres into the silicon at a moderate concentration, phosphorus is the best choice. Phosphorus is an n-type dopant that diffuses deeper and wider than arsenic or antimony, which makes it suitable for forming a deep well during a long drive-in, and its high solid solubility easily meets the moderate concentration requirement. This is exactly the combination used in the BICMOS5 process: arsenic for the SN source/drain regions and phosphorus for the N-Well.

3.9 Process Simulation - 12

In the masking simulations, a $0.4 \mu\text{m}$ layer of photoresist, oxide or nitride is deposited on the silicon, after which a 160 keV phosphorus implantation is performed. In the resulting plots the negative part of the depth axis is the masking layer and the positive part the silicon.

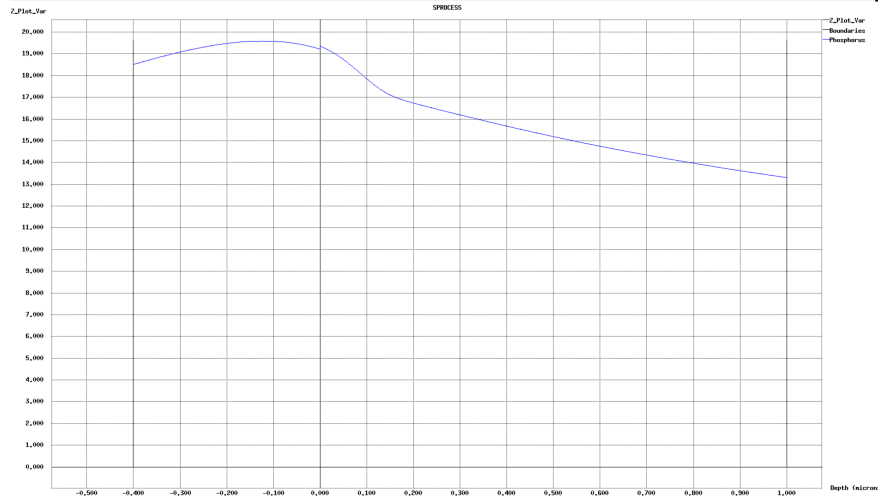


Figure 12: Phosphorus implantation profile through a $0.4 \mu\text{m}$ photoresist mask. The vertical axis is $\log_{10}$ of the phosphorus concentration in cm^{-3} ; negative depths lie inside the resist.

Figure 12 shows that the phosphorus peak lies inside the resist, but the tail of the profile crosses the silicon surface at a concentration near 10^{19} cm^{-3} and stays above the 10^{16} cm^{-3} level down to more than $0.3 \mu\text{m}$ into the silicon. A $0.4 \mu\text{m}$ photoresist layer is therefore not a sufficient implantation mask for this phosphorus implantation.

3.10 Process Simulation - 13

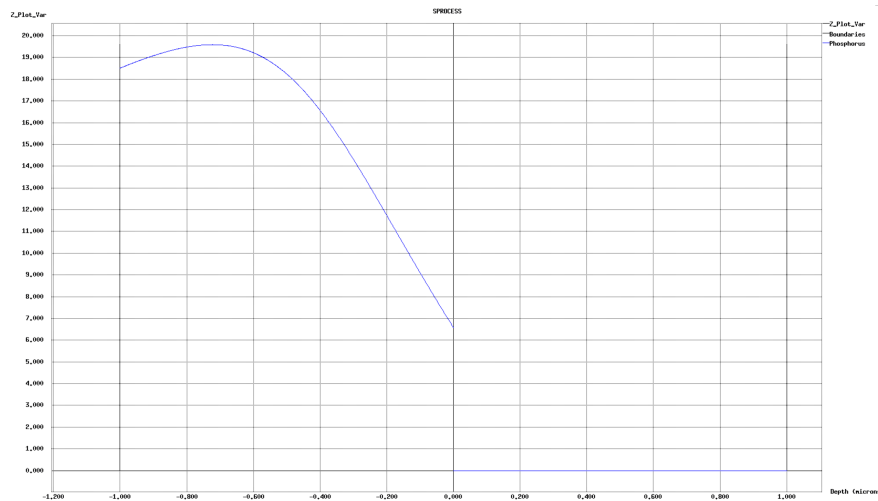


Figure 13: Phosphorus implantation profile through a $1 \mu\text{m}$ photoresist mask. The vertical axis is $\log_{10}$ of the phosphorus concentration in cm^{-3} ; negative depths lie inside the resist.

After increasing the thickness of the photoresist layer from $0.4 \mu\text{m}$ to $1 \mu\text{m}$, the whole phosphorus profile is contained inside the resist and the concentration drops to zero well before the silicon interface. Therefore, a $1 \mu\text{m}$ photoresist layer is sufficient as an implantation mask for this phosphorus implantation: resist can mask, but it needs to be roughly 2.5 times thicker than the denser hard-mask materials below.

3.11 Process Simulation - 14 & 15

For the oxide mask (Figure 14), the phosphorus peak lies deeper inside the mask, at about $-0.2 \mu\text{m}$, and only a weak tail reaches the silicon: the concentration at the silicon surface is below 10^{14} cm^{-3} , i.e. at or

below the substrate background doping. A $0.4\ \mu\text{m}$ oxide layer therefore just barely masks this implantation, but with essentially no process margin.

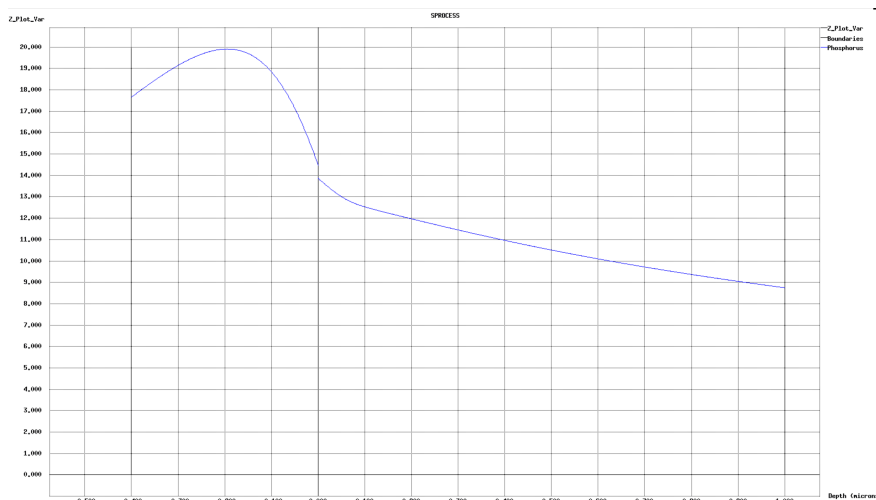


Figure 14: Phosphorus implantation profile through a $0.4\ \mu\text{m}$ oxide mask. The vertical axis is $\log_{10}$ of the phosphorus concentration in cm^{-3} ; negative depths lie inside the oxide.

For the nitride mask (Figure 15), the phosphorus concentration already drops to zero about $0.03\ \mu\text{m}$ before the silicon interface, so the silicon stays completely clean. In terms of stopping power, a $0.4\ \mu\text{m}$ nitride mask is therefore the most effective of the three, followed by oxide and then photoresist. This ordering follows the film densities: nitride ($3.1\ \text{g}/\text{cm}^3$) is denser than oxide ($2.2\ \text{g}/\text{cm}^3$), which is denser than resist (about $1.2\ \text{g}/\text{cm}^3$), and a denser film stops the ions over a shorter distance.

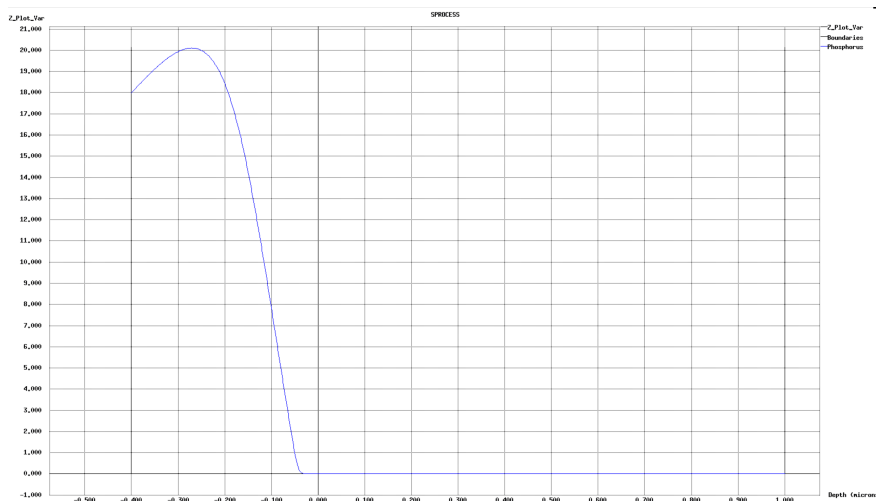


Figure 15: Phosphorus implantation profile through a $0.4\ \mu\text{m}$ nitride mask. The vertical axis is $\log_{10}$ of the phosphorus concentration in cm^{-3} ; negative depths lie inside the nitride.

Nitride is almost never used as a standard implantation mask because a nitride layer is deposited at high temperatures, which means it is hard to process. In addition, it is difficult to pattern and remove. Because of its high intrinsic tensile stress, it can also introduce stress in the wafer.

In standard processing, photoresist is used most often because it is easy to pattern. Although photoresist has relatively low stopping power, it can be made thick enough to absorb the ions so that none reach the silicon. Finally, it can be removed easily after implantation.

4 Device Simulations

In the device-simulation part, the complete BICMOS5 process flow was first simulated in two dimensions with Sentaurus Process (assignments 1 to 3, which involve loading and running the provided Sentaurus Workbench project), after which the electrical behaviour of the resulting NMOS and PMOS structures was computed with Sentaurus Device. The device simulator solves the Poisson equation and the electron and hole continuity equations on the 2D mesh, including Shockley-Read-Hall and Auger recombination and doping- and field-dependent mobility, and sweeps the terminal voltages to produce I_D-V_G and I_D-V_{DS} characteristics. The questions below follow the assignment numbering of the manual.

4.1 Device Simulation - 4

The 2D cross-section of the NMOS generated by SVisual (Figure 16) generally matches the reference cross-section from the manual: the n^+ source and drain regions sit in the p-type substrate with the substrate contacts at the edges and the aluminium contacts on top. The main difference is that the manual draws the source, drain, substrate contact, oxide and metal regions as ideal rectangles, while the simulated structure shows a more realistic picture: the doped regions have rounded corners from lateral diffusion during the anneals, and the doping concentration changes gradually with depth instead of having a sharp boundary.

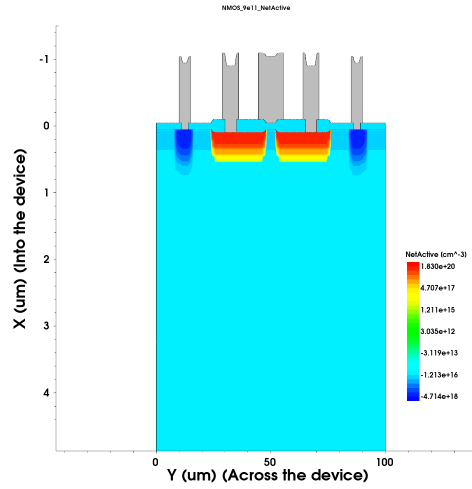


Figure 16: Simulated NMOS cross-section showing the net active dopant concentration for a V_T -adjust dose of $9 \times 10^{11} \text{ cm}^{-2}$.

The 2D cross-section of the PMOS (Figure 17) also matches the reference drawing: it is the mirror image of the NMOS, with p^+ source and drain regions inside the deep n-type well instead of n^+ regions in the p-substrate, and it shows the same rounded corners and graded junction transitions.

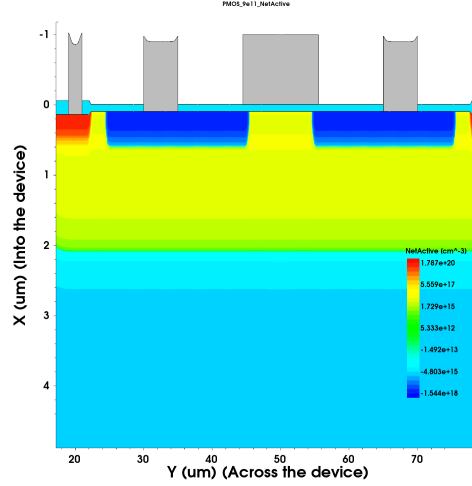


Figure 17: Simulated PMOS cross-section showing the net active dopant concentration for a V_T -adjust dose of $9 \times 10^{11} \text{ cm}^{-2}$.

4.2 Device Simulation - 5

The simulated profiles are net active doping profiles, i.e. they show the $|n - p|$ concentration, and the doping type of each region follows from which species dominates. In the NMOS, the source and drain are n-type SN regions, while the channel and substrate are p-type. In the PMOS, the device is located inside the n-type N-Well, and the source and drain are p-type SP regions; below the well the wafer returns to the p-type substrate.

The simulated junction depths were extracted from the exported cutline data by finding numerically where the signed net active doping crosses zero. The extracted values are $0.537 \mu\text{m}$ for SN, $2.085 \mu\text{m}$ for the N-Well, and $0.594 \mu\text{m}$ for SP.

The simulated values differ from the calculated values of Section 2 because the analytical calculations assume purely Gaussian profiles and neglect all thermal steps after the implantation, while the simulation includes the full process flow and the dopant redistribution during every anneal. The SN junction is simulated about six times deeper than calculated: the analytical estimate ignores the gate-oxidation anneal at 1000°C that diffuses the arsenic further, and the solid solubility limit clips the As peak from the theoretical $1.75 \times 10^{21} \text{ cm}^{-3}$ down to the simulated $1.8 \times 10^{20} \text{ cm}^{-3}$, which pushes the junction deeper for the same dose. The SP junction is about three times deeper than calculated for the same reason, amplified by the fast interstitialcy diffusion of boron in the implantation-damaged silicon. The N-Well, in contrast, is simulated slightly shallower than calculated (within about 13%): there the analytical limited-source model is a good approximation, because the 4-hour drive-in dominates the profile and smooths the implant into a nearly Gaussian shape. Table 4 summarizes the comparison.

Table 4: Junction depths: analytical calculation (Section 2) versus Sentaurus simulation.

Implantation	Calc. junction depth (μm)	Simulated junction depth (μm)
SN	0.087	0.54
N-WELL	2.37	2.09
SP	0.186	0.59

4.3 Device Simulation - 6

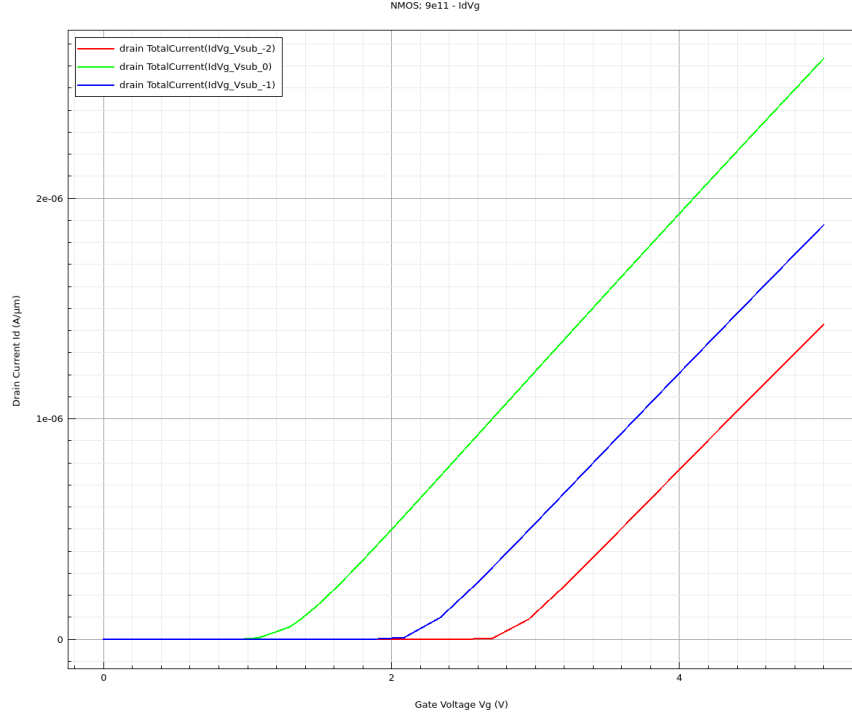


Figure 18: Simulated NMOS I_D - V_G characteristics at $V_D = 0.1$ V for substrate bias $V_{\text{Sub}} = 0, -1$, and -2 V (V_T -adjust dose $9 \times 10^{11} \text{ cm}^{-2}$).

The threshold voltage is extracted from the exported I_D - V_G data with the constant-current method, defining V_T as the gate voltage at which $|I_D| = 100$ nA. As a cross-check, linear extrapolation at maximum transconductance was applied as well: in the linear region (small V_D) the drain current is

$$I_D = \mu_n C_{\text{ox}} \frac{W}{L} \left(V_{GS} - V_T - \frac{V_{DS}}{2} \right) V_{DS},$$

which is linear in V_{GS} , so extrapolating the tangent taken at the point of maximum transconductance $g_{m,\text{max}} = \max(\Delta I_D / \Delta V_{GS})$ to $I_D = 0$ gives an intercept $V_{GS0} = V_T + V_{DS}/2$ from which V_T follows. Both methods agree to better than 0.1 V for the NMOS.

The constant-current extraction yields $V_T = 1.385, 2.339$ and 2.966 V for $V_{\text{Sub}} = 0, -1$ and -2 V, respectively (linear extrapolation gives 1.31, 2.23 and 2.84 V).

The threshold voltage increases as V_{Sub} becomes more negative. This is the body effect: a more negative V_{Sub} raises the reverse source-body bias $V_{SB} = -V_{\text{Sub}}$, widening the channel-substrate depletion region and adding depletion charge that the gate must balance before inversion. This matches the theory of the body effect [3],

$$V_T = V_{T0} + \gamma \left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F} \right), \quad (6)$$

where V_{T0} is the threshold at zero body bias, $\gamma = \sqrt{2q\epsilon_s N_A} / C_{\text{ox}}$ the body-effect coefficient and $2\phi_F$ the surface potential at inversion. Fitting Equation (6) to the three simulated points with $2\phi_F \approx 0.7$ V gives $\gamma \approx 2.0 \text{ V}^{1/2}$, consistent within a few percent for both bias points, which corresponds to a channel doping in the low 10^{17} cm^{-3} range, a plausible value for the surface concentration after the V_T -adjust implant.

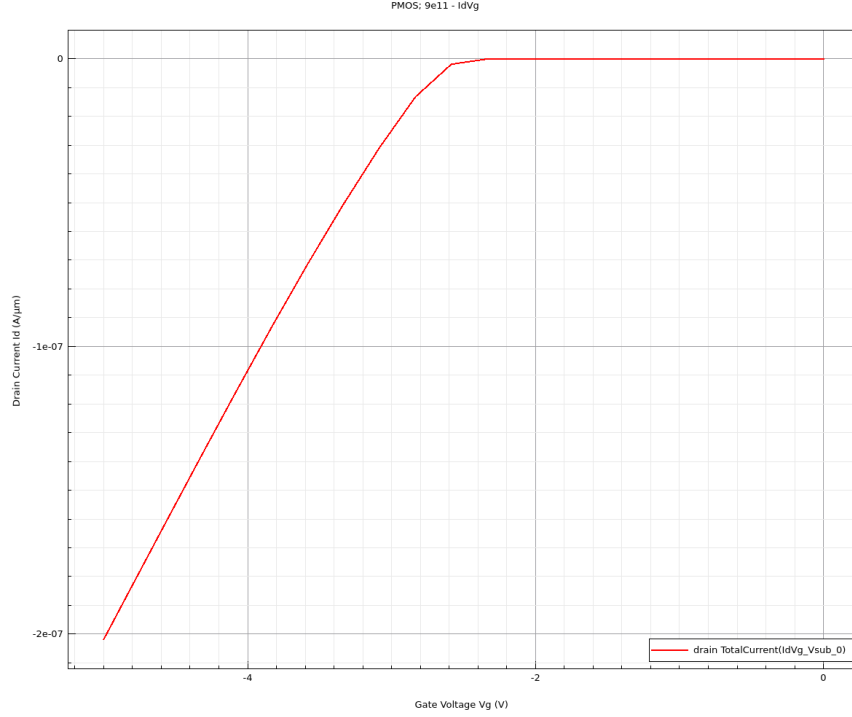


Figure 19: Simulated PMOS I_D - V_G characteristics at $V_D = -0.1$ V for substrate bias $V_{\text{Sub}} = 0$ V (V_T -adjust dose $9 \times 10^{11} \text{ cm}^{-2}$).

The PMOS is simulated only at $V_{\text{Sub}} = 0$ V. The constant-current method gives a threshold voltage of $V_T = -3.91$ V (linear extrapolation is unreliable here because of the sharp curvature of the I_D - V_G curve near threshold). This value is quite negative for a CMOS process: even the highest V_T -adjust dose of $9 \times 10^{11} \text{ cm}^{-2}$ does not compensate the N-Well surface enough to bring the PMOS threshold close in magnitude to the NMOS one.

4.4 Device Simulation - 7

Figure 20 shows one curve per gate voltage, so V_G is the parameter that changes from line to line. The curves for $V_G \leq 1$ V stay near zero because the device is below threshold ($V_T \approx 1.39$ V). Each conducting curve first rises almost proportionally to V_{DS} (linear region), then flattens once V_{DS} passes the knee at $V_{DS} \approx V_G - V_T$ (saturation region). The knee shifts to higher V_{DS} as V_G increases; for $V_G = 6$ and 7 V it falls close to the edge of the swept range, so those curves are still rising at $V_{DS} = 5$ V and do not fully saturate.

The saturation currents follow the long-channel square law $I_{D,\text{sat}} \propto (V_G - V_T)^2$ to first order: the gap between successive curves widens with V_G , as the quadratic dependence predicts. A closer look shows a mild compression at the highest gate voltages: going from $V_G = 3$ to 6 V roughly doubles the overdrive, so the square law predicts a factor of about 8 in current, while the simulation gives about a factor of 7. This slight sub-quadratic behaviour is the onset of velocity saturation, where the carrier drift velocity begins to saturate at high lateral fields and $I_{D,\text{sat}}$ tends towards a linear dependence on the overdrive. The saturated curves also show a small positive slope, i.e. weak channel-length modulation. Both effects are mild here, as expected for a long-channel device; they return much more strongly in the measured $1 \mu\text{m}$ devices of Section 6.

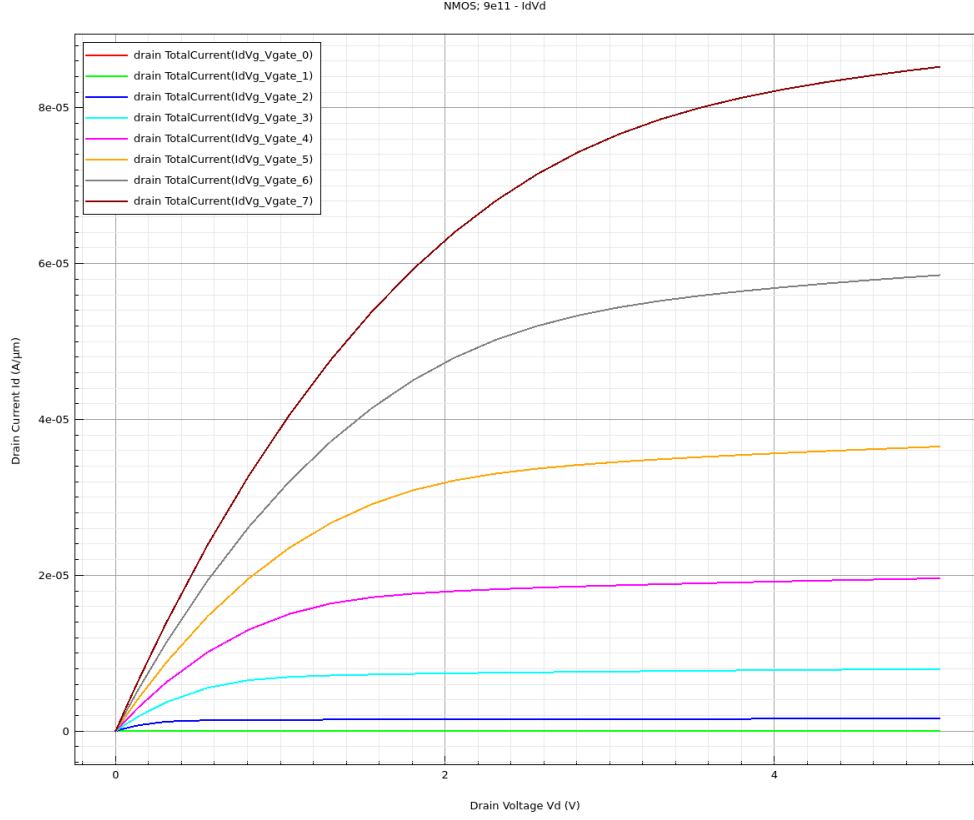


Figure 20: Simulated NMOS I_D - V_{DS} characteristics for $V_G = 0$ to 7 V in 1 V steps at $V_{\text{Sub}} = 0$ (V_T -adjust dose $9 \times 10^{11} \text{ cm}^{-2}$).

The PMOS characteristics in Figure 21 show the same behaviour, mirrored in sign. The linear and saturation regions and the square-law spacing match the NMOS case. At equal $|V_G|$ the PMOS currents are smaller in magnitude than the NMOS currents, because the hole mobility is about a third of the electron mobility and the larger PMOS threshold magnitude ($V_T \approx -3.91 \text{ V}$) leaves less gate overdrive; the curves for $|V_G| \leq 3 \text{ V}$ carry essentially no current because the device is below threshold. The velocity-saturation compression is also less pronounced than in the NMOS, since the lower hole mobility keeps the carriers further from their saturation velocity at the same fields.

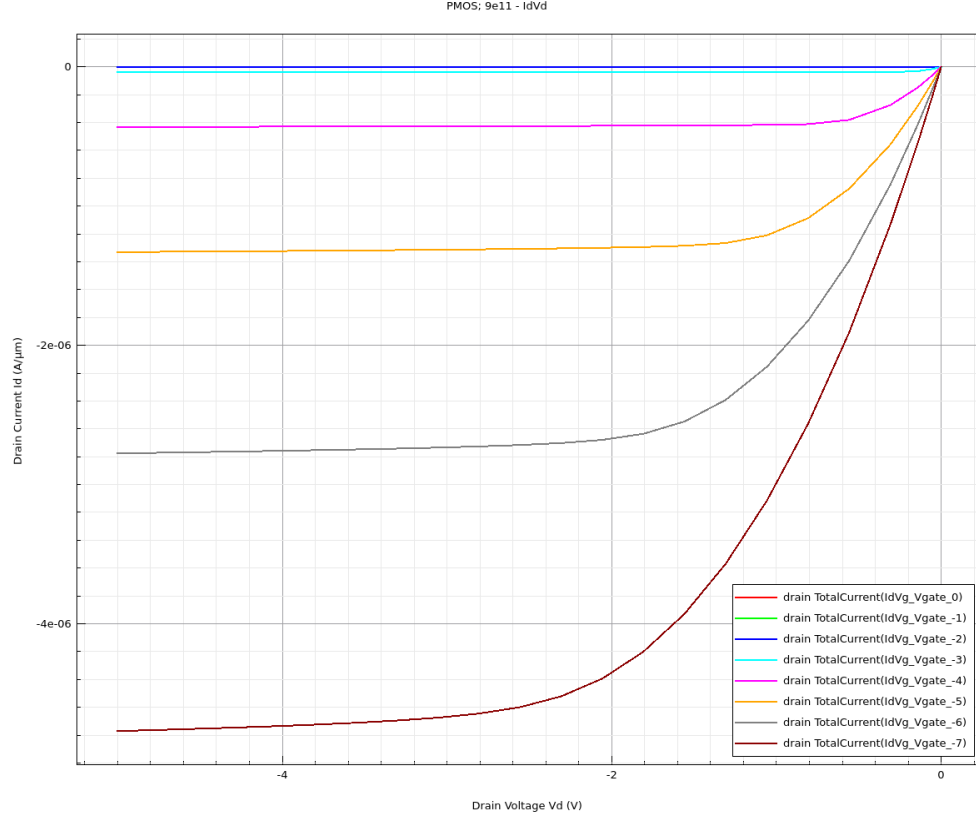


Figure 21: Simulated PMOS I_D - V_{DS} characteristics for $V_G = 0$ to -7 V in 1 V steps at $V_{Sub} = 0$ (V_T -adjust dose $9 \times 10^{11} \text{ cm}^{-2}$).

4.5 Device Simulation - 9

After lowering the V_T -adjust dose from 9×10^{11} to $3 \times 10^{11} \text{ cm}^{-2}$ and re-running the process simulation, the two doses give almost identical results for the 2D cross-section, the vertical cut through the source, and the horizontal cut along the device; the only visible difference is in the vertical doping profile through the channel. This is expected, since every step that forms the source, drain and N-Well runs before the V_T -adjust implantation, so changing the dose only affects the shallow boron layer at the channel surface. The channel cuts of both devices at the two doses are shown in Figures 22a to 22d.

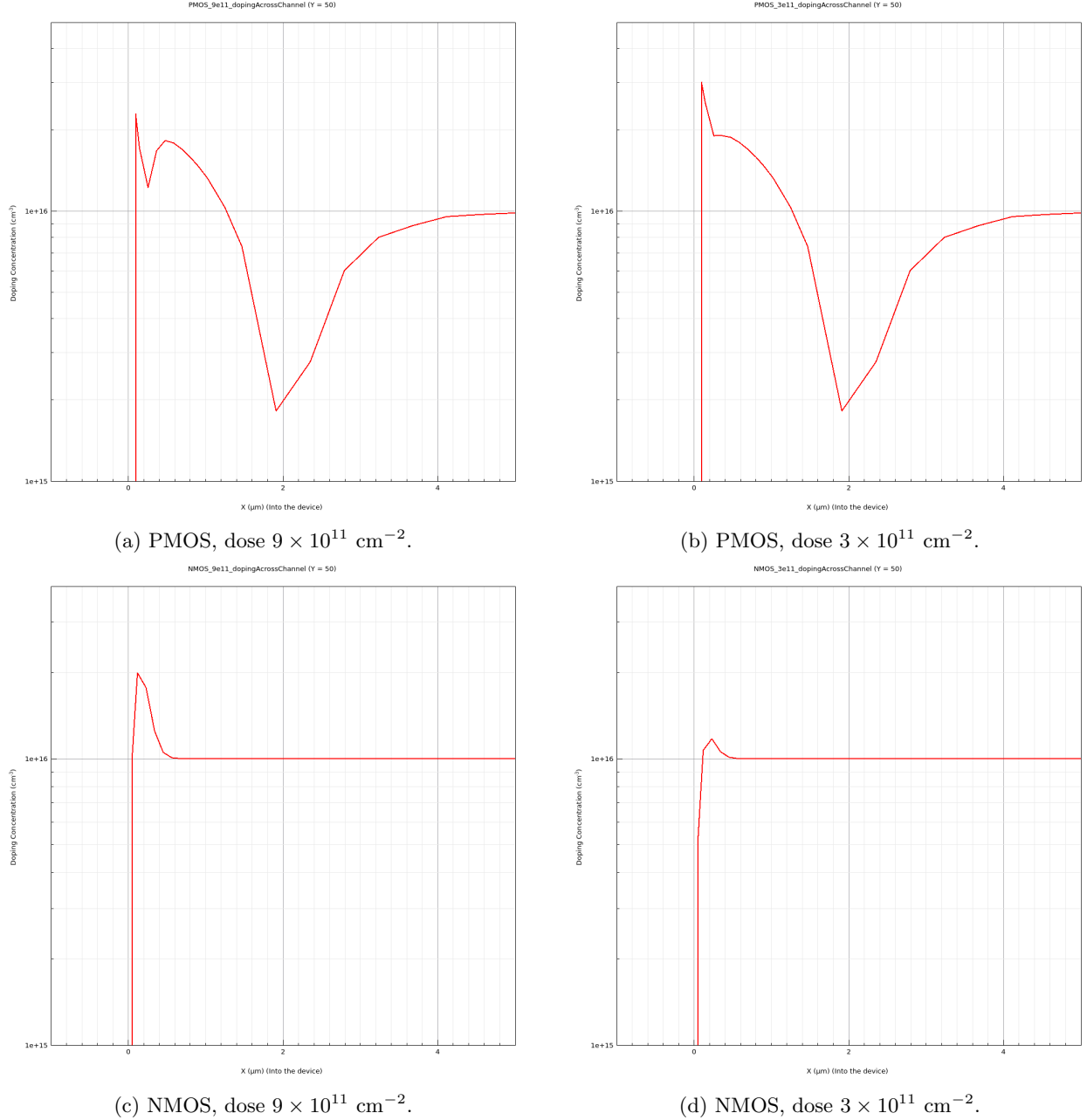


Figure 22: Simulated net dopant concentration across the channel of the PMOS and NMOS for the two V_T -adjust doses.

For the PMOS, the boron V_T -adjust is implanted into the n-type well, where it compensates the phosphorus donors near the surface. At $9 \times 10^{11} \text{ cm}^{-2}$ the net doping shows a clear compensation notch just below the surface, where the boron most closely balances the phosphorus; at $3 \times 10^{11} \text{ cm}^{-2}$, one third of the dose, the boron only dents the profile and the notch is absent. Below about $0.5 \mu\text{m}$ the boron is no longer present and the two profiles coincide.

For the NMOS, the boron is implanted into the p-type substrate, so instead of compensating a well it adds directly to the channel acceptors. The net doping rises to a peak just below the surface: about $2 \times 10^{16} \text{ cm}^{-3}$ at $9 \times 10^{11} \text{ cm}^{-2}$, well above the $1 \times 10^{16} \text{ cm}^{-3}$ substrate background, versus only about $1.2 \times 10^{16} \text{ cm}^{-3}$ at $3 \times 10^{11} \text{ cm}^{-2}$. The excess above the background scales with the implanted dose, and below about $0.5 \mu\text{m}$ both profiles return to the substrate level.

4.6 Device Simulation - 10 & 11

The I_D - V_G simulations were repeated at the two lower V_T -adjust doses, and the threshold voltages were extracted with the same constant-current method ($|I_D| = 100$ nA). Table 5 gives the resulting overview requested by the manual. The cells at $V_{\text{Sub}} = -1$ and -2 V for the lower doses were not required (the manual marks them X to save simulation time), but our sweep produced them as well, so they are included in parentheses; the PMOS is simulated at $V_{\text{Sub}} = 0$ V only.

Table 5: Simulated threshold voltage at the three V_T -adjust doses, extracted with the constant-current method ($V_D = 0.1$ V for NMOS, -0.1 V for PMOS). Values in parentheses were not required by the manual.

V_T -adjust dose	V_T (V) for 3E11	V_T (V) for 6E11	V_T (V) for 9E11
NMOS, $V_{\text{Sub}} = 0$ V	0.905	1.162	1.385
NMOS, $V_{\text{Sub}} = -1$ V	(1.695)	(2.013)	2.339
NMOS, $V_{\text{Sub}} = -2$ V	(2.264)	(2.622)	2.966
PMOS, $V_{\text{Sub}} = 0$ V	-4.372	-4.372	-3.907

Two trends stand out. First, the NMOS threshold grows almost linearly with the dose (about 0.08 V per 10^{11} cm^{-2}), because the boron adds directly to the acceptor concentration of the lightly doped p-type channel. Second, the PMOS threshold is completely insensitive at the two lower doses and only starts to move at $9 \times 10^{11} \text{ cm}^{-2}$. This is consistent with the doping profiles of Device Simulation 9: the boron sheet at the lower doses corresponds to a concentration two orders of magnitude below the donor concentration at the N-Well surface, so it cannot shift the net surface charge noticeably. Only when the boron starts to compensate a significant fraction of the surface donors does the PMOS threshold respond. Both trends confirm the qualitative V_T -adjust theory of Section 2: a single boron implantation moves both threshold voltages in the positive direction, raising $|V_T|$ of the NMOS and lowering $|V_T|$ of the PMOS. Finally, the body-effect columns show that the threshold shift with substrate bias grows with dose, as expected from the body-effect coefficient $\gamma \propto \sqrt{N_A}$: more boron means a higher channel doping and a stronger body effect. These simulated thresholds are compared with the measured ones in Section 6.

5 Processing Assignments

The processing part of the course took place in the class-100 cleanroom of the Else Kooi Laboratory. Three groups of assignments were carried out: in-line sheet resistance measurements on the implanted monitor wafers to determine the doping concentrations, oxide deposition with thickness and etch-rate measurements, and the patterning of the aluminium interconnect layer of a BICMOS5 wafer by wet and dry etching.

5.1 Doping Concentration

The sheet resistance of the three implanted monitor wafers was measured with the CDE ResMap four-point probe over a ten-point scan. The results, including the average value, the standard deviation and the spread across the wafer, are listed in Table 6.

Table 6: Sheet resistance of the N-Well, SN and SP wafers, measured with the four-point probe equipment.

Wafer	R_s ($\Omega/\square$)	Std. dev. ($\Omega/\square$)	Range ($\Omega/\square$)
N-Well	1488.8	10.9	1471–1507
SN	57.4	0.99	55.7–58.6
SP	510.6	2.4	506.6–513.9

The ordering of the three values follows directly from the doping levels: the lightly doped N-Well has the highest sheet resistance even though it is the deepest layer, the arsenic SN layer is doped up to the solid solubility limit and has the lowest, and the boron SP layer sits in between because boron reaches a lower active concentration and a lower mobility than arsenic.

The junction depths of the N-Well, SN and SP obtained from the simulations in Section 4 are 2.085, 0.537 and 0.594 μm respectively. The sheet resistance and the junction depth together give the depth-averaged resistivity of each layer:

$$R_s = \frac{\rho}{x_j} \implies \rho = R_s x_j. \quad (7)$$

The average active dopant concentration N is then read from the Irvin curve for the corresponding carrier type at the calculated resistivity [1]. The results are listed in Table 7.

Table 7: Average resistivity and average active dopant concentration from the Irvin curves.

Layer	$\rho = R_s x_j$ (Ωcm)	Type	N (cm^{-3})
N-Well	0.31	n	$N_D \approx 1.5 \times 10^{16}$
SN	3.1×10^{-3}	n	$N_D \approx 2 \times 10^{19}$
SP	3.0×10^{-2}	p	$N_A \approx 1.5 \times 10^{18}$

These dopant concentration values are averaged over the diffused profile, so they fall below the simulated peak concentrations from Section 3, as expected: the SN average sits about an order of magnitude below its solid-solubility-clipped peak of $1.8 \times 10^{20} \text{cm}^{-3}$, which is normal for a steeply graded junction, while the flatter N-Well and SP profiles give averages close to the simulated levels. These three concentrations are used again for comparison with the Van der Pauw measurements in Section 6.

5.2 Oxide Deposition, Thickness Measurements and Etch Rate

Two silicon wafers received an oxide layer: one grown by wet (thermal) oxidation and one deposited by PECVD from TEOS at a low temperature of 350 to 400 $^\circ\text{C}$. The thickness of both oxide layers was measured optically at five points per wafer, before and after a 20 s dip in buffered HF (BHF, 1:7). The measured oxide thicknesses and the uniformity of both depositions are listed in Table 8; the range column gives the spread (maximum minus minimum) over the five measurement points.

Table 8: Oxide thickness and uniformity before and after 20 s etching in BHF (1:7).

Method	Before etch		After etch		Removed (nm)
	Thickness (nm)	Range (nm)	Thickness (nm)	Range (nm)	
Wet oxidation	294.34	1.54	260.23	8.09	34.1
PECVD	324.94	5.65	214.34	20.25	110.6

Before etching, the oxide layer produced by wet oxidation is more uniform than that produced by PECVD. The oxide thickness produced by wet oxidation varies by 1.54 nm across the wafer, while the PECVD film varies by 5.65 nm, roughly four times as much. This is because thermal oxidation is rate-limited by oxidant diffusion through the already-grown oxide: it grows more slowly where the layer is thicker, which levels the layer. In contrast, the PECVD thickness follows the gas flow and plasma density patterns in the chamber, which vary from point to point.

After etching, the uniformity became worse for both wafers, and worse for PECVD than for wet oxidation. The spread grew from 1.54 to 8.09 nm for wet oxidation, and from 5.65 to 20.25 nm for PECVD.

The etch rate is calculated from the removed thickness during the 20 s etch in buffered HF:

$$r_{\text{wet}} = \frac{34.1}{20} = 1.71 \text{ nm/s}, \quad r_{\text{PECVD}} = \frac{110.6}{20} = 5.53 \text{ nm/s}, \quad (8)$$

which shows that the PECVD oxide is etched about 3.2 times faster than the thermal oxide. The thermal value matches the literature BHF (1:7) etch rate for thermal SiO_2 of roughly 1.5 to 2 nm/s. The difference comes from film density. The wet oxide is grown above 1000 $^\circ\text{C}$ by consuming the silicon itself, so it is dense, stoichiometric and fully bonded. The PECVD oxide is deposited at low temperature from TEOS: it is less dense and more porous, and it still holds Si-OH groups and water. BHF moves through this open film

quickly and attacks the weaker bonds, so the etch runs faster and less evenly. A densification anneal after deposition would lower the PECVD etch rate towards the thermal value, which is the usual industrial fix.

Therefore, compared to PECVD deposition, wet oxidation gives a dense and uniform oxide layer. Its main disadvantage is that it operates at a high temperature, which adds to the thermal budget and drives further diffusion of the dopants already in the wafer. It also consumes silicon as it grows, and it only grows on exposed silicon, so it cannot be used over metal.

PECVD, on the other hand, operates at a low temperature, can be deposited over metal and other finished layers, and coats any surface. Its disadvantage is the lower film quality: the oxide is less dense and more porous, which leads to a less uniform layer with more defects and trapped charge.

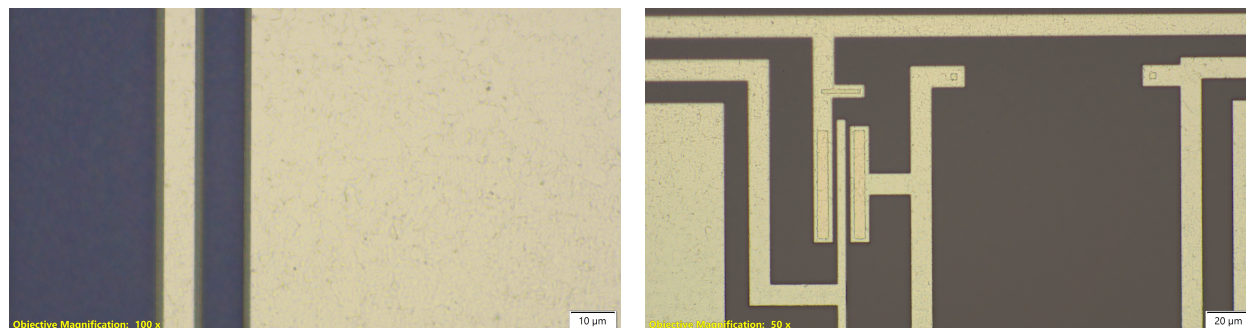
For the gate oxide, wet oxidation is the right choice. The gate dielectric sits directly under the channel, so interface-trap density, leakage and breakdown all matter, and only a grown thermal oxide gives a clean Si/SiO₂ interface. PECVD is more suitable for layers where temperature is the limiting constraint, such as inter-metal dielectric and passivation over aluminium. Among thermal oxides, dry oxidation is preferred over wet for the gate because it is denser and more controllable, but between the two methods used here wet oxidation is clearly the one to use.

5.3 CMOS Wafer Metal Etching and Etching Selectivity

Two BICMOS5 wafers processed up to the 0.2 μm aluminium layer were used to pattern the IC mask. One wafer was etched wet in phosphoric acid at 35 °C, the other dry in a Cl/HBr plasma. Both were inspected under the optical microscope; representative images are shown in Figure 23.

The pattern after lithography shows the contact openings and metal lines lining up with the underlying devices, so the lithography was good enough to proceed to the etch.

After the aluminium is etched away (visible as a colour change from shiny to grey), the surface looks rough and grainy rather than smooth and reflective. This roughness comes from the 1% silicon in the aluminium alloy, which is added to prevent spiking: phosphoric acid etches the aluminium but leaves the silicon behind as small grains. The subsequent poly-Si dip in HNO₃-HF removes them.



(a) Wet etch, metal lines at 100 $\times$.

(b) Dry etch, comb structure at 50 $\times$.

Figure 23: Microscope images of the aluminium pattern after wet etching (left) and dry etching (right). The wet-etched lines are narrowed and rounded by the isotropic underetch and the exposed field oxide keeps its blue/violet colour, while the dry-etched lines hold their drawn width with straight edges over a brown/grey (thinned) field oxide.

The wet etch is isotropic: the acid etches sideways under the resist at the same rate as downwards, so the metal edges become rounded and less defined and the lines end up visibly narrower than drawn (Figure 23a).

The dry etch is anisotropic: the directional ion bombardment etches mainly downwards rather than sideways, so the edges stay straight and even the narrow comb lines keep a clean, well-defined shape at their full mask width (Figure 23b).

The field oxide started at approximately 100 nm. In the wet-etched wafer the field is blue/violet, the interference colour of about 100 nm of SiO₂, so the oxide is essentially untouched. Phosphoric acid does not attack oxide, so with the 0.2 μm of aluminium removed against at most a few nanometres of oxide loss, the aluminium-to-oxide selectivity is very high, effectively better than 100:1.

In the dry-etched wafer the field is brown/grey, the interference colour of a clearly thinner oxide, so

a substantial part of the 100 nm SiO_2 has been removed. The plasma erodes SiO_2 as well as aluminium through its physical bombardment component, so the selectivity is far lower, on the order of 10:1 to 20:1.

Wet etching is therefore the more selective process, since it removes almost no oxide. Its disadvantages are the poor dimensional control caused by the sideways undercut, which rules out fine-pitch patterns, and the need for good resist adhesion to avoid delamination during the underetch. Dry etching gives straight, accurately sized lines at the cost of eroding the underlying oxide, and it requires careful endpoint control; in addition, the plasma hardens the photoresist, which must then be removed in an oxygen plasma instead of acetone. This is the classic trade-off between selectivity and dimensional control in wet versus dry processing.

6 Measurements

During the final part of the course, electrical measurements were performed on the process control module (PCM) structures of the BICMOS5 wafer: Van der Pauw (VdP) squares, electrical line width measurement (ELM) resistors, and NMOS/PMOS transistors in all four V_T -adjust quadrants. These measurements close the loop of the course: they are compared with the analytical calculations of Section 2, the simulations of Sections 3 and 4, and the in-line processing measurements of Section 5.

The measurements were taken on a Cascade 31 automatic probe station with a Keysight B1500A parameter analyzer controlled by IC-CAP, using four-point probing throughout, so that probe and contact resistances drop out of the results. Following the course guidelines, no IC-CAP screenshots are used here: all raw data were exported and re-plotted in MATLAB.

The junction depths x_j of the N-Well, SN and SP implants were taken from the Sentaurus simulations in Section 4, and the in-line sheet resistances from Section 5, so the measurements here can be compared directly with the earlier process and simulation results.

6.1 The BICMOS5 Test Wafer

The course wafer is a 10 cm wafer processed at the Else Kooi Laboratory, containing over 160 dies of 6×6 mm [1]. The wafer is divided into four quarters, each of which received a different boron V_T -adjust dose during processing: 0 (top left), 3×10^{11} (bottom left), 6×10^{11} (top right) and $9 \times 10^{11} \text{ cm}^{-2}$ (bottom right), as shown in Figure 24(a). Measuring the same device type in each quarter therefore reveals the influence of the V_T -adjust dose directly.

Each die (Figure 24(b)) contains several blocks, of which four are used in this course: a PMOS transistor block and an NMOS transistor block with various gate geometries, a block of Van der Pauw structures, and the electrical linewidth measurement structures. The resistive structures are strips implanted into a region of the opposite doping type, so each strip is isolated from its surroundings by a pn-junction; this isolation only holds as long as that junction stays reverse biased, a condition that returns in the choice of the measurement voltages below.

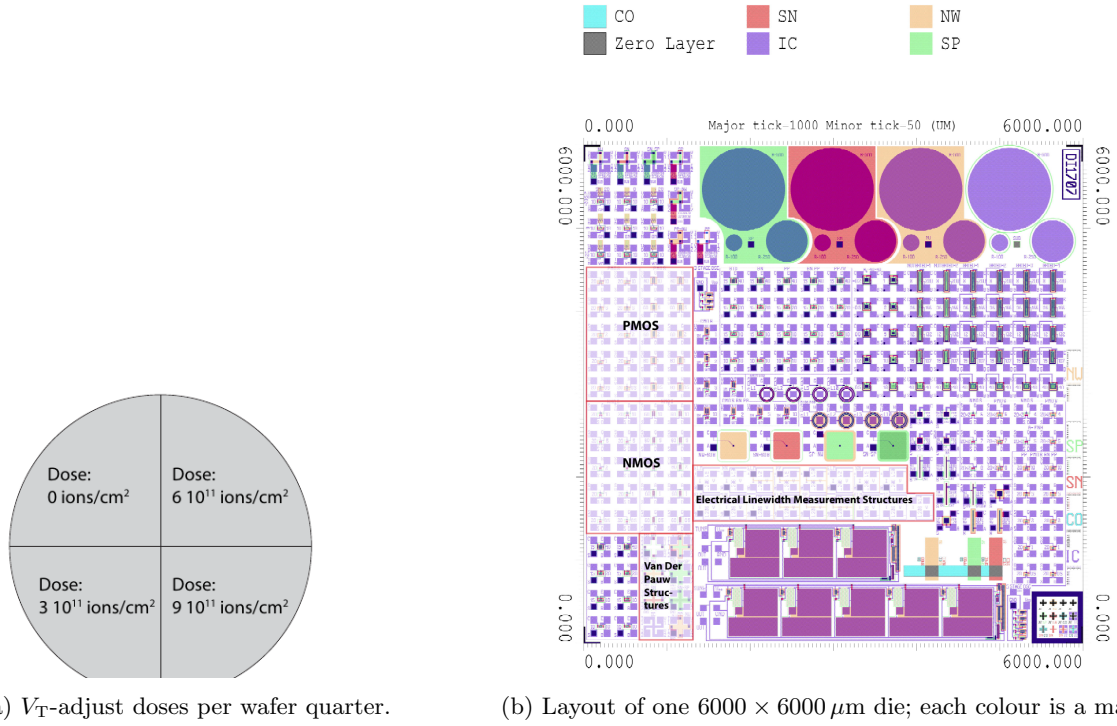


Figure 24: The BICMOS5 test wafer: (a) the four V_T -adjust quarters and (b) the die layout with the PMOS, NMOS, ELM and Van der Pauw blocks used in this course [1].

6.2 Van der Pauw Sheet Resistance and Doping Concentration

The sheet resistance $R_s = \rho/d$ of a layer with resistivity ρ and thickness d allows a designer to compute the resistance of any strip of that layer as $R = R_s L/W$, with L the strip length and W its width. The Van der Pauw structures on the die are greek crosses: a $30 \times 30 \mu\text{m}$ square of the layer to be measured with a contact on each side, drawn for each implantation and layer combination (Figure 25). The VdP structures were measured in the top-left quadrant (V_T -adjust dose 0) using a symmetric four-point configuration. The analyzer swept the force voltage between -0.2V and $+0.2\text{V}$ at grounded substrate, and the resistance $R_{AB,CD}$ was obtained from the least-squares slope of the sensed voltage difference $V_1 - V_2$ versus the forced current I_{force} . For a symmetric VdP structure a single measurement suffices, and the sheet resistance follows from the Van der Pauw relation [5]

$$R_s = \frac{\pi}{\ln 2} R_{AB,CD} \approx 4.532 R_{AB,CD}, \quad (9)$$

where $R_{AB,CD}$ is the resistance measured by forcing a current through contacts A and B and sensing the voltage over contacts C and D, and the factor $\pi/\ln 2$ corrects for the radial current flow in the square structure.

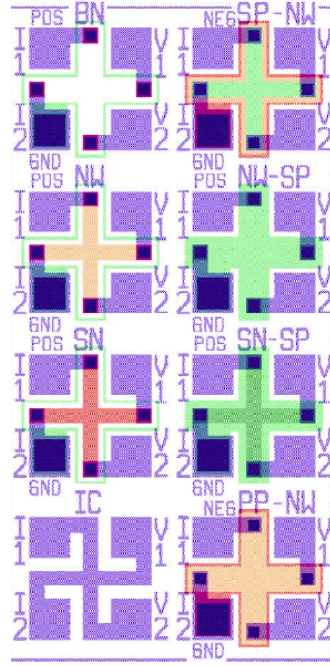


Figure 25: Greek-cross Van der Pauw structures for the different layers and layer combinations, with the current (I) and voltage (V) probe pads indicated [1].

The small sweep range is essential: the implanted layers form pn-junctions with the substrate (or well) underneath, and a silicon junction only conducts appreciably above its exponential knee of about 0.6 V. Within ± 0.2 V the junction current is orders of magnitude below the lateral sheet current for either sweep polarity, so the carriers stay confined in the implanted layer, as the VdP method requires. At larger voltages the forward-biased half of the sweep would open the junction and current would spread through the entire substrate, collapsing the measured resistance.

The measured I/V sweeps and their linear fits are shown in Figure 26; the extracted values are summarized in Table 9. The last column repeats the in-line monitor values from Section 5 for comparison.

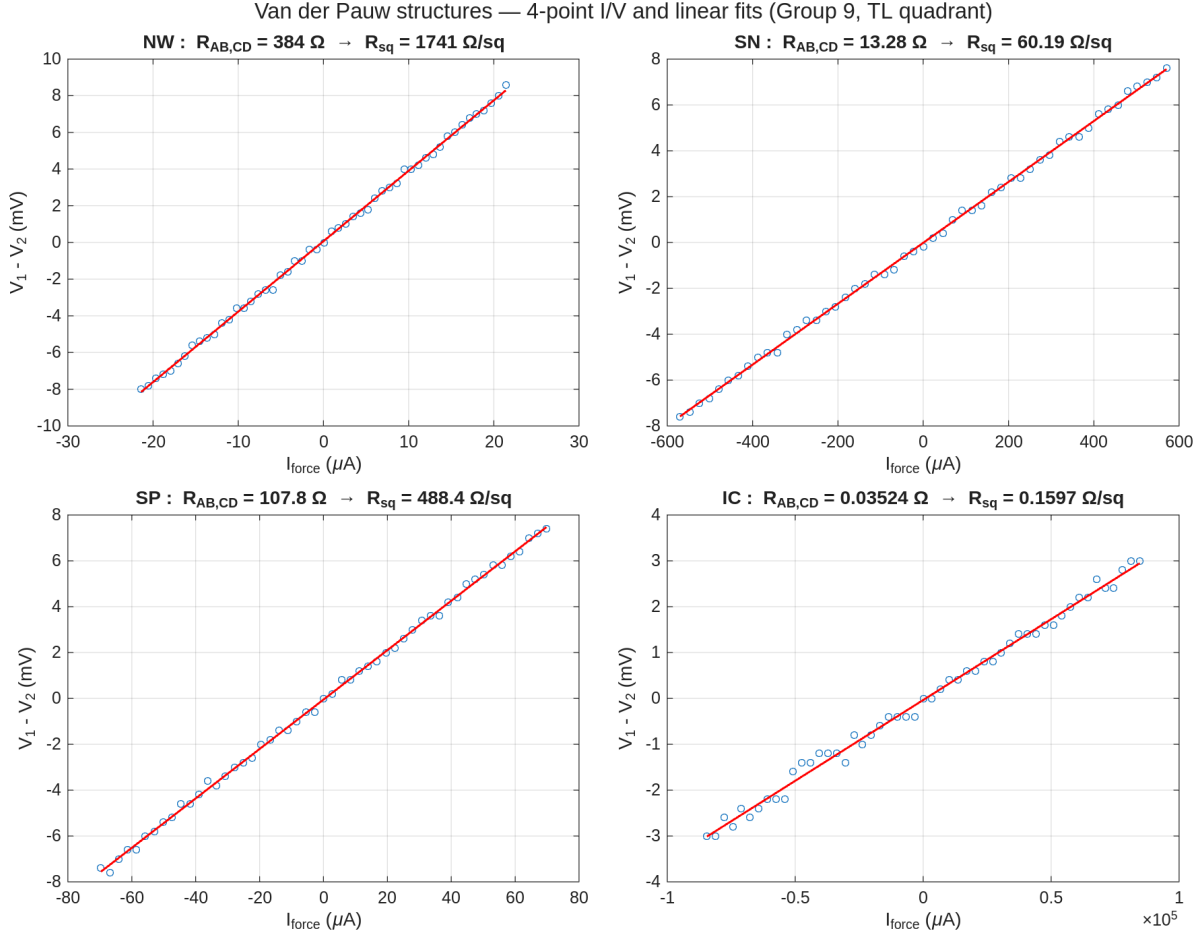


Figure 26: Van der Pauw four-point I/V sweeps and linear fits for the N-Well, SN, SP and IC (metal) layers, measured in the top-left quadrant. All fits have $R^2 > 0.99$.

Table 9: Van der Pauw sheet resistances of the implanted layers and metal, compared to the in-line monitor values from Section 5.

Layer	$R_{AB,CD}$ (Ω)	R_s ($\Omega/\square$)	Fit R^2	In-line R_s ($\Omega/\square$)
N-Well	384.02	1740.5	0.9991	1488.8
SN	13.28	60.19	0.9991	57.4
SP (in N-Well)	107.75	488.4	0.9990	510.6
IC metal	0.0352	0.160	0.9939	—

The SN and SP values agree with the in-line monitor resistances to within about 5%, while the N-Well shows a larger deviation of roughly 17%. This is consistent with the N-Well being the most lightly doped layer, and therefore the most sensitive to wafer position, surface depletion, and the difference between the blanket monitor wafer used in the cleanroom and the drawn VdP cloverleaf on the PCM.

The IC metal has a sheet resistance of only $0.160 \Omega/\square$, three to four orders of magnitude lower than the implanted silicon layers, whose values range from about $60 \Omega/\square$ for SN to $1740 \Omega/\square$ for the N-Well. With the known film thickness $d = 0.2 \mu\text{m}$, the metal resistivity follows as $\rho = R_s d = 3.2 \times 10^{-6} \Omega\text{cm}$, which matches the literature value for Al with 1% Si. The root cause of the difference in magnitude is the free-carrier density: aluminium contains roughly one conduction electron per atom (about 10^{23}cm^{-3}), whereas the implanted layers only contain 10^{16} to 10^{19}cm^{-3} activated dopants. The higher carrier mobility

in lightly doped silicon only partly offsets this. For this reason the metal layer is used for low-resistance interconnects, while the implanted layers are suitable for making resistors.

Using the simulated junction depths x_j from Section 4, the depth-averaged resistivity of each implanted layer follows from

$$\rho = R_s x_j. \quad (10)$$

The average active dopant concentration $\bar{N}$ is then obtained from the Irvin curves for the corresponding carrier type, here evaluated with the Masetti doping-dependent mobility model [6]. The resulting resistivities and concentrations are given in Table 10.

Table 10: Depth-averaged resistivity and active dopant concentration from the VdP measurements and Irvin curves, compared to the values from Section 5.

Layer	x_j (μm)	$\rho = R_s x_j$ (Ωcm)	Type	$\bar{N}$ (cm^{-3})	Sec. 5 N (cm^{-3})
N-Well	2.085	0.363	n	1.5×10^{16}	1.5×10^{16}
SN	0.537	3.2×10^{-3}	n	2.3×10^{19}	2×10^{19}
SP	0.594	2.9×10^{-2}	p	1.6×10^{18}	1.5×10^{18}

The doping concentrations obtained from the three routes are therefore consistent within their meaning. The VdP and in-line values agree to within about 10% for all three layers, which is remarkable for two independent electrical measurements on different structures. Compared to the simulations of Section 3, the extracted values are depth-averaged active concentrations: for the steep, solid-solubility-limited SN profile the average sits roughly an order of magnitude below the simulated peak of $1.8 \times 10^{20} \text{cm}^{-3}$, while for the flatter N-Well and SP profiles the average is close to the simulated levels. It should also be noted that the measured sheet resistances are only consistent with the simulated junction depths: inserting the much shallower analytical depths of Section 2 into Equation (10) would produce unphysical concentrations, which confirms that the analytical model misses the thermal budget of the real process.

6.3 Electrical Line Width Measurements and Lateral Out-Diffusion

During the anneals, dopant atoms not only diffuse deeper into the silicon but also sideways out of the implantation window, so a resistor line is electrically wider than drawn. To probe this lateral out-diffusion, ELM structures with drawn widths $W = 2, 5$ and $10 \mu\text{m}$ and length $L = 206 \mu\text{m}$ were measured for the N-Well and SN implants. Each ELM structure is a differential resistor with four contacts (Figure 27): the outermost contacts force a current through the strip, and two inner contacts sense the voltage drop over the $206 \mu\text{m}$ length, so the measurement is again free of contact resistance. For each resistor a four-point measurement was taken and the apparent sheet resistance was computed as

$$R_{s,\text{app}} = R \frac{W}{L}, \quad (11)$$

using the drawn width W . The results are listed in Table 11 and plotted against the drawn width in Figure 28.

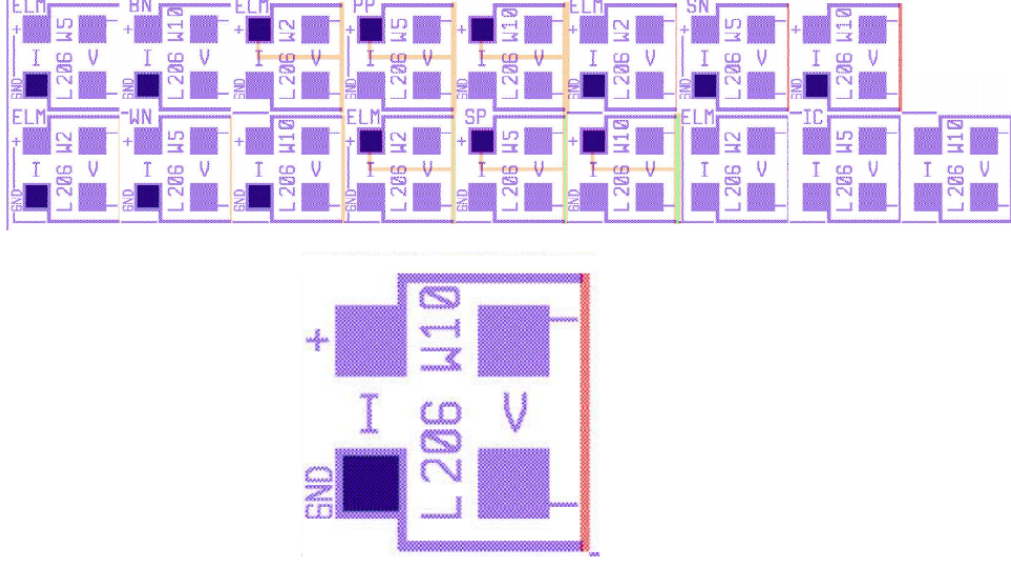


Figure 27: The ELM block with the differential resistors of all layers (top) and the detailed view of one structure (bottom). The two capital letters give the mask (e.g. SN), the number after W the drawn line width in μm ; all strips have $L = 206 \mu\text{m}$ [1].

Table 11: Raw ELM resistances and apparent sheet resistances for N-Well and SN resistors ($L = 206 \mu\text{m}$).

Layer	W (μm)	R (Ω)	$R_{s,\text{app}}$ ($\Omega/\square$)
N-Well	2	58832.7	571.2
N-Well	5	38478.1	933.9
N-Well	10	19689.2	955.8
SN	2	5244.8	50.9
SN	5	2273.5	55.2
SN	10	1171.1	56.8

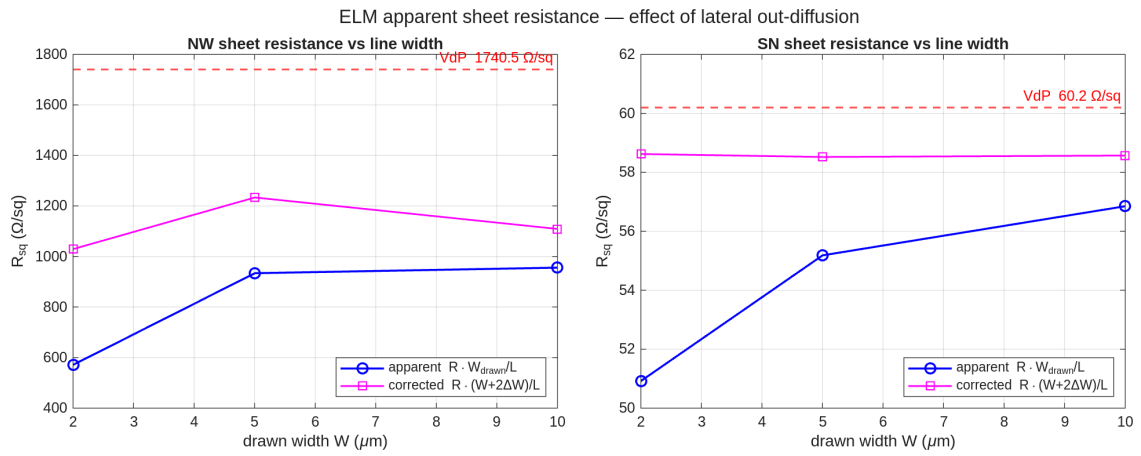


Figure 28: Apparent ELM sheet resistance versus drawn line width for the N-Well and SN implants. The dashed lines indicate the Van der Pauw reference values.

For both implants the apparent sheet resistance computed with the drawn width is systematically lower

than the VdP value, and the deficit grows as the line becomes narrower (-67% for the $2\mu\text{m}$ N-Well line versus -15% for the $2\mu\text{m}$ SN line). This shows that the conducting strip is wider than its nominal geometry because of lateral dopant diffusion; a fixed widening matters relatively more for narrow lines. The effective width can be written as

$$W_{\text{eff}} = W + 2\Delta W, \quad (12)$$

where ΔW is the lateral out-diffusion per side.

Assuming a constant ΔW , the conductance is linear in the drawn width, $1/R = (W + 2\Delta W)/(R_s L)$, so a straight-line fit of $1/R$ versus W yields the true sheet resistance from the slope and ΔW from the intercept. The fits are shown in Figure 29 and the extracted values are summarized in Table 12.

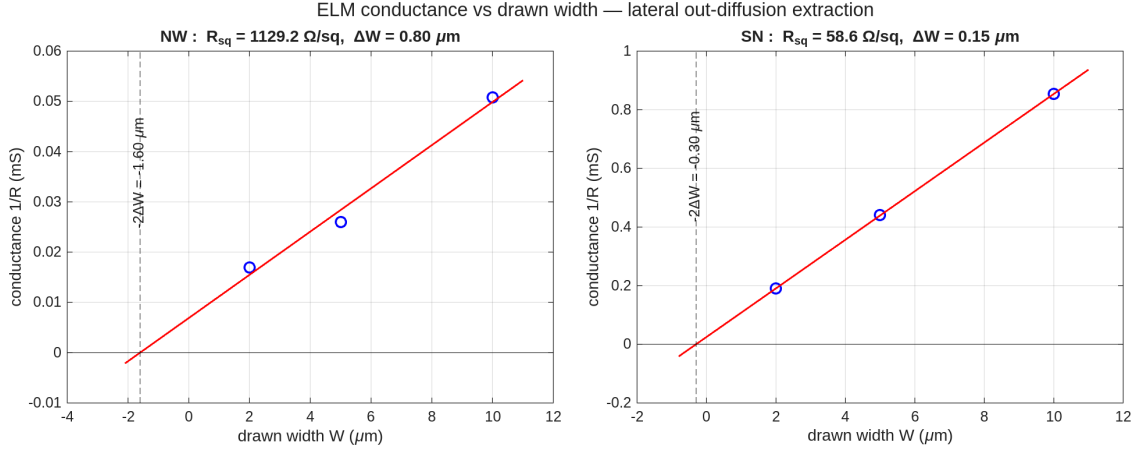


Figure 29: ELM conductance versus drawn width with linear fits. The extrapolation to zero conductance yields the lateral out-diffusion ΔW per side.

Table 12: Sheet resistance and lateral out-diffusion extracted from the ELM structures.

Layer	$R_{\text{s,ELM}} (\Omega/\square)$	$\Delta W (\mu\text{m})$	Fit R^2
N-Well	1129.2	0.80	0.986
SN	58.6	0.15	0.999999

For the SN implant the extraction is textbook-like: the conductance is perfectly linear in width, the ELM sheet resistance agrees with the VdP value ($60.2 \Omega/\square$) to within about 3%, and the extracted lateral out-diffusion is only $\Delta W \approx 0.15 \mu\text{m}$ per side. The N-Well behaves differently: the three points are not exactly collinear and the free fit returns a sheet resistance well below the VdP value. Anchoring the sheet resistance to the VdP value ($1740 \Omega/\square$) instead, the narrow lines consistently indicate an out-diffusion closer to $2.1 \mu\text{m}$ per side, comparable to the vertical junction depth. The constant- ΔW model is simply too crude for the N-Well, whose laterally diffused edges are graded over about $2 \mu\text{m}$, so each line width samples a different effective sheet resistance.

The large difference in out-diffusion between the two implants has two causes. First, the thermal budget: the N-Well phosphorus receives a 4 h drive-in at 1150°C , which is what makes it $2 \mu\text{m}$ deep, and diffusion is isotropic, so the dopant spreads sideways by a comparable amount. The SN arsenic only sees the much shorter gate-oxidation anneal. Second, the species: arsenic diffuses roughly an order of magnitude slower than phosphorus at the same temperature. Both effects together explain why ΔW for the N-Well is more than five times the SN value.

6.4 Transistor Characterisation: Threshold, Mobility, Short-Channel Effects

The PMOS and NMOS blocks of the die each contain 20 transistors with different combinations of the two gate dimensions, labelled in the layout by the device type letter with the two dimensions in μm next

to it (Figure 30). NMOS and PMOS transistors with gate geometries $W:L = 20:1, 20:2, 20:5, 20:10$ and $60:2$ (gate width : gate length, in μm) were measured in the bottom-right quadrant (V_T -adjust dose $9 \times 10^{11} \text{ cm}^{-2}$). Transfer characteristics $I_D(V_G)$ were recorded at $|V_D| = 0.1 \text{ V}$ and threshold voltages and mobilities were extracted using the long-channel unified model. In the linear region ($|V_{DS}|$ small) the drain current is

$$I_D = \mu C_{\text{ox}} \frac{W}{L} \left(V_{\text{GS}} - V_T - \frac{V_{\text{DS}}}{2} \right) V_{\text{DS}}, \quad (13)$$

where μ is the carrier mobility, $C_{\text{ox}} = \epsilon_0 \epsilon_{\text{SiO}_2} / t_{\text{ox}} = 3.45 \times 10^{-8} \text{ F/cm}^2$ the oxide capacitance per unit area for the $t_{\text{ox}} = 100 \text{ nm}$ gate oxide, W and L the gate width and length, and V_{GS} , V_{DS} and V_T the gate-source, drain-source and threshold voltages. The threshold voltage was extracted by linear extrapolation of $I_D(V_G)$ at the point of maximum transconductance $g_{\text{m,max}}$ (corrected by $V_{\text{DS}}/2$), and the mobility follows from the transconductance as

$$\mu = \frac{g_{\text{m,max}} L}{C_{\text{ox}} W |V_{\text{DS}}|}. \quad (14)$$

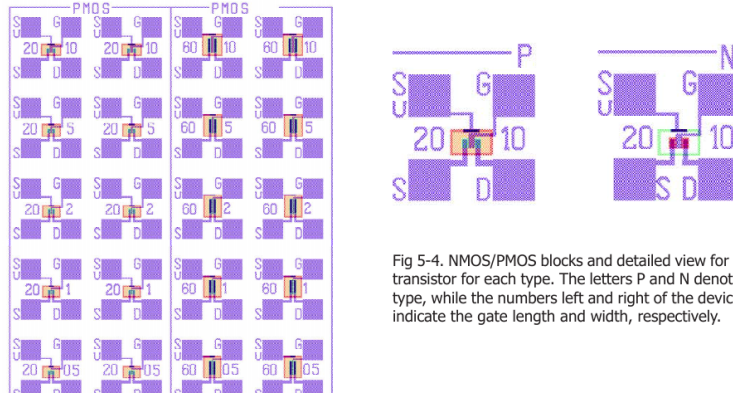


Fig 5-4. NMOS/PMOS blocks and detailed view for a transistor for each type. The letters P and N denote type, while the numbers left and right of the device indicate the gate length and width, respectively.

Figure 30: Part of the PMOS transistor block and the detailed view of one PMOS and one NMOS device with their source (S), gate (G), drain (D) and substrate (SU) pads; the numbers next to each device give its gate dimensions in μm [1].

Figure 31 shows the measured transfer characteristics of all geometries; the extracted parameters are listed in Table 13.

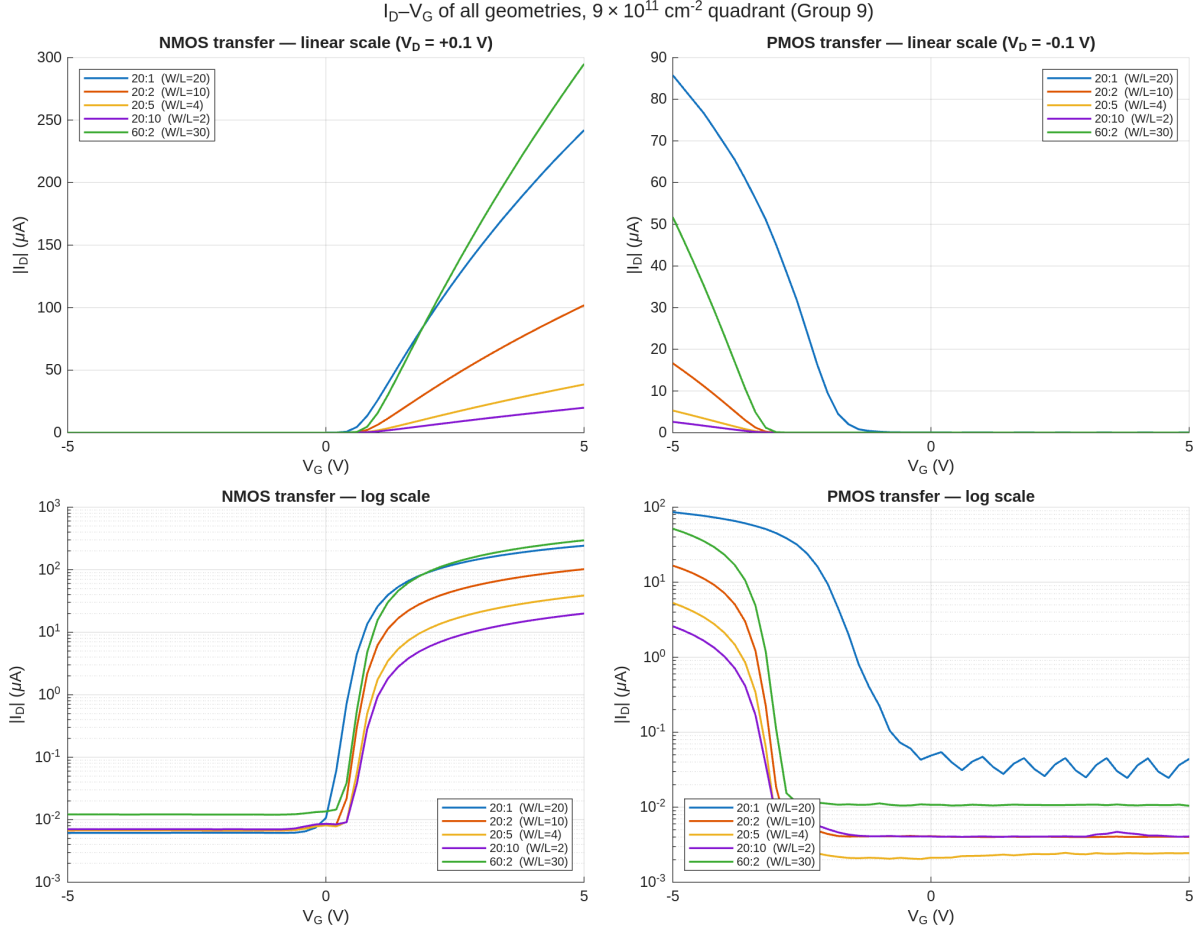


Figure 31: Measured NMOS and PMOS transfer characteristics for all gate geometries at $|V_D| = 0.1 \text{ V}$ in the $9 \times 10^{11} \text{ cm}^{-2}$ quadrant, on linear and logarithmic current scales.

Table 13: Threshold voltages and mobilities extracted from the transfer characteristics in the $9 \times 10^{11} \text{ cm}^{-2}$ quadrant ($V_D = \pm 0.1 \text{ V}$).

Type	$W:L$	V_T (V)	$g_{m,\max}$ (μS)	μ ($\text{cm}^2/\text{V s}$)
NMOS	20:1	0.572	68.0	985
NMOS	20:2	0.740	27.5	796
NMOS	20:5	0.819	10.1	734
NMOS	20:10	0.811	5.18	750
NMOS	60:2	0.779	80.6	778
PMOS	20:1	-1.733	39.0	565
PMOS	20:2	-3.262	10.4	301
PMOS	20:5	-3.300	3.25	236
PMOS	20:10	-3.305	1.59	230
PMOS	60:2	-3.213	31.5	304

The influence of the geometry on the characteristics is exactly what Equation (13) predicts: the on-current and the transconductance scale with W/L ($g_{m,\max}$ runs from 5.2 to $80.6 \mu\text{S}$ as W/L goes from 2 to 30), while the shape of the curves, including the threshold and the subthreshold slope, is unchanged for $L \geq 2 \mu\text{m}$. Threshold voltage and mobility are process parameters and should not depend on geometry,

and for gate lengths $L \geq 2 \mu\text{m}$ they indeed remain constant to within a few percent, as Figure 32 shows. The long-channel devices give electron mobilities around 750 to $800 \text{ cm}^2/\text{Vs}$ and hole mobilities around 230 to $300 \text{ cm}^2/\text{Vs}$, a ratio of about three, as expected from the electron-to-hole mobility ratio in silicon. Both values sit below the bulk mobilities (1417 and $470 \text{ cm}^2/\text{Vs}$ [3]), which is expected for surface-channel conduction with the V_T -adjust implant located at the surface.

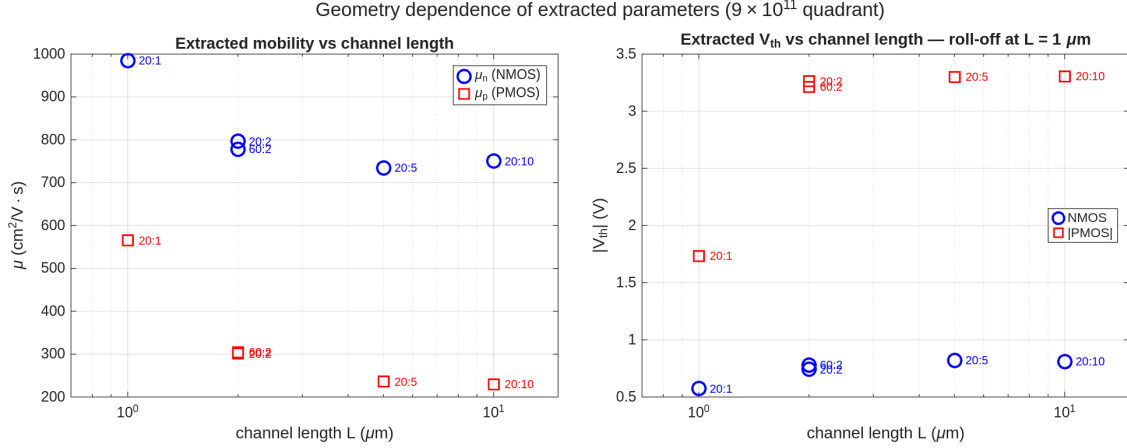


Figure 32: Extracted mobility and threshold voltage versus gate length. Both parameters are flat for $L \geq 2 \mu\text{m}$ and break away at $L = 1 \mu\text{m}$.

At the shortest gate length $L = 1 \mu\text{m}$ (the 20:1 devices), clear short-channel effects appear. The NMOS threshold voltage rolls off from about 0.82 V to 0.57 V , because the source and drain depletion regions support part of the channel depletion charge. The extracted mobility is artificially inflated (985 versus about $750 \text{ cm}^2/\text{Vs}$), reflecting effective channel shortening and the onset of drain-induced barrier lowering (DIBL) rather than a true change in carrier transport. The PMOS suffers far more, with the threshold moving from about -3.30 V to -1.73 V , because its junctions are deeper ($0.59 \mu\text{m}$) and sit in the lightly doped N-Well, which makes the depletion regions wide compared to the $1 \mu\text{m}$ channel.

6.4.1 Threshold Voltage versus V_T -Adjust Dose

To quantify the effect of the boron V_T -adjust implant, the $W:L = 20:5$ transistors were measured in all four quadrants, with doses 0 , 3×10^{11} , 6×10^{11} and $9 \times 10^{11} \text{ cm}^{-2}$. Figure 33 shows the transfer curves per quadrant, and Figure 34 compares the extracted threshold voltages with the Sentaurus simulations from Section 4. The values are listed in Table 14.

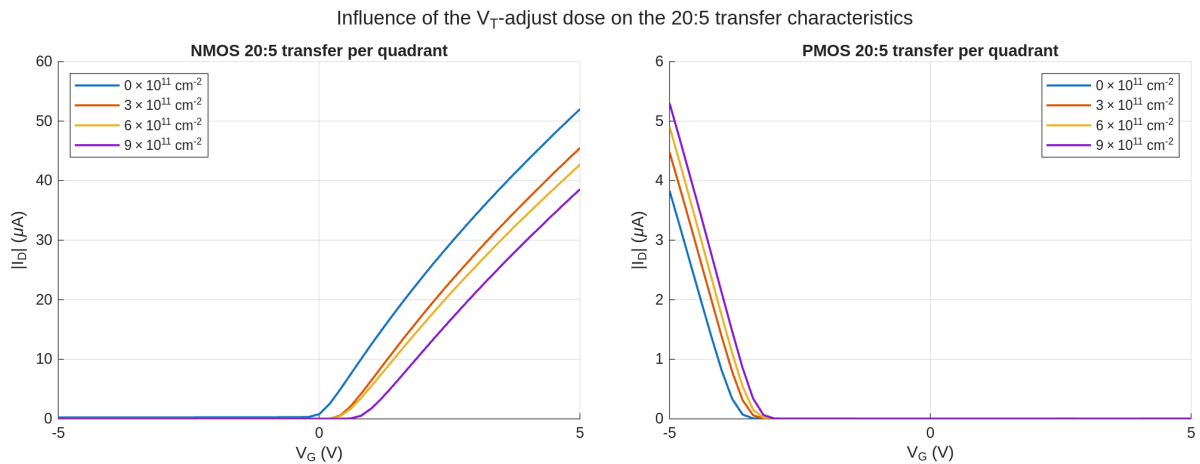


Figure 33: Transfer characteristics of the 20:5 transistors in the four wafer quadrants (V_T -adjust doses 0 to $9 \times 10^{11} \text{ cm}^{-2}$).

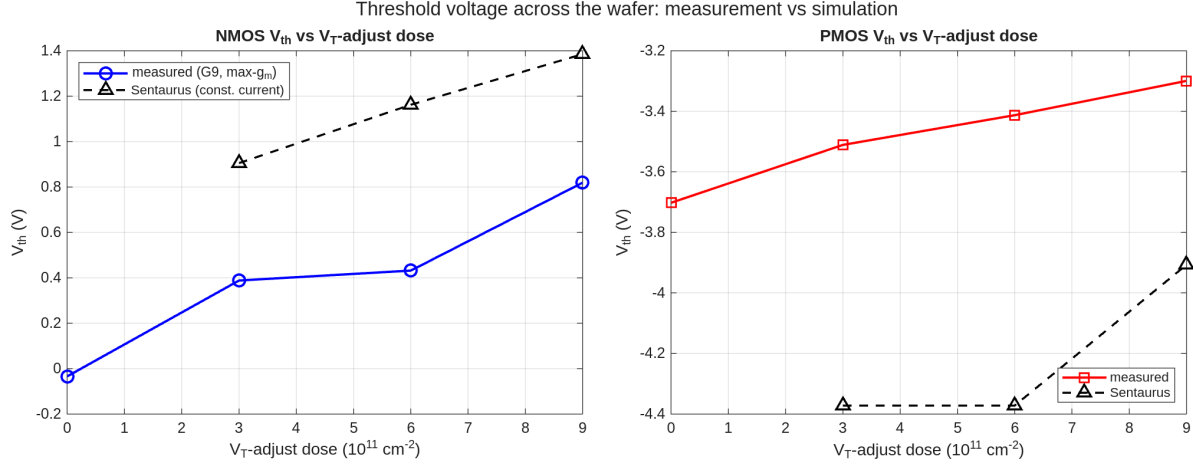


Figure 34: Measured threshold voltage versus V_T -adjust dose for the 20 : 5 NMOS and PMOS transistors, compared with the Sentaurus device simulations (constant-current extraction).

Table 14: Threshold voltage versus V_T -adjust dose for the 20 : 5 transistors, measured and simulated (Section 4).

Dose (cm^{-2})	Meas. $V_{T,\text{NMOS}}$ (V)	Meas. $V_{T,\text{PMOS}}$ (V)	Sim. $V_{T,\text{NMOS}}$ (V)	Sim. $V_{T,\text{PMOS}}$ (V)
0	-0.034	-3.702	—	—
3×10^{11}	0.388	-3.511	0.905	-4.372
6×10^{11}	0.432	-3.413	1.162	-4.372
9×10^{11}	0.819	-3.300	1.385	-3.907

The NMOS threshold voltage increases monotonically with the boron dose, from nearly 0 V (an almost normally-on device) to about 0.82 V, because the additional acceptor charge in the channel depletion region must be imaged by a larger gate voltage. The same boron implant partially compensates the N-Well surface underneath the PMOS gate, making its threshold voltage less negative by roughly 0.4 V over the same dose range. This is precisely the mechanism predicted by the theory in Section 2: a single unmasked boron implant shifts both thresholds in the positive direction, which increases $|V_T|$ for the NMOS and decreases it for the PMOS.

The comparison with the simulations shows the same trend and a similar dose sensitivity, but with an offset: the simulated NMOS values sit about 0.5 V higher and the simulated PMOS values about 0.6 V more negative than the measurements. Part of this offset is methodological, since the simulated values use a constant-current criterion while the measured values use maximum-transconductance extrapolation, and part is calibration, since the simulation does not include the oxide and interface charge Q_{ox} of the real gate stack, which shifts the flat-band voltage. The NMOS is roughly four times more dose-sensitive than the PMOS because the boron lands directly in its channel depletion region, whereas for the PMOS it must compete with the much higher donor concentration of the N-Well surface.

Comparing the two devices overall: the trends are the same with opposite signs, but the absolute performance differs strongly. The PMOS threshold magnitude is about four times the NMOS value, because the channel sits in the doped N-Well and the maximum applied boron dose is nowhere near enough for a symmetric pair, and the PMOS drive current and transconductance are about three times lower at equal geometry because of the lower hole mobility. A circuit designer in this process must size the PMOS about three times wider than the NMOS, and a symmetric CMOS process would need a larger V_T -adjust dose.

6.4.2 Output Characteristics and Channel-Length Modulation

Output characteristics $I_D(V_{\text{DS}})$ were measured for the 20 : 5 and 20 : 1 devices in the $9 \times 10^{11} \text{ cm}^{-2}$ quadrant, with the gate voltage stepped through the linear and saturation regions ($V_G = 1$ to 7 V for NMOS,

–1 to –8 V for PMOS). The measured families of curves are shown in Figures 35 and 36. In saturation the unified model gives

$$I_D = \frac{\mu C_{\text{ox}} W}{2 L} (V_{\text{GS}} - V_T)^2 [1 + \lambda (V_{\text{DS}} - V_{\text{DS,sat}})], \quad (15)$$

where λ is the channel-length modulation parameter and $V_{\text{DS,sat}}$ the saturation voltage. Ideally the saturation current depends only on the gate overdrive; real devices show a finite output slope.

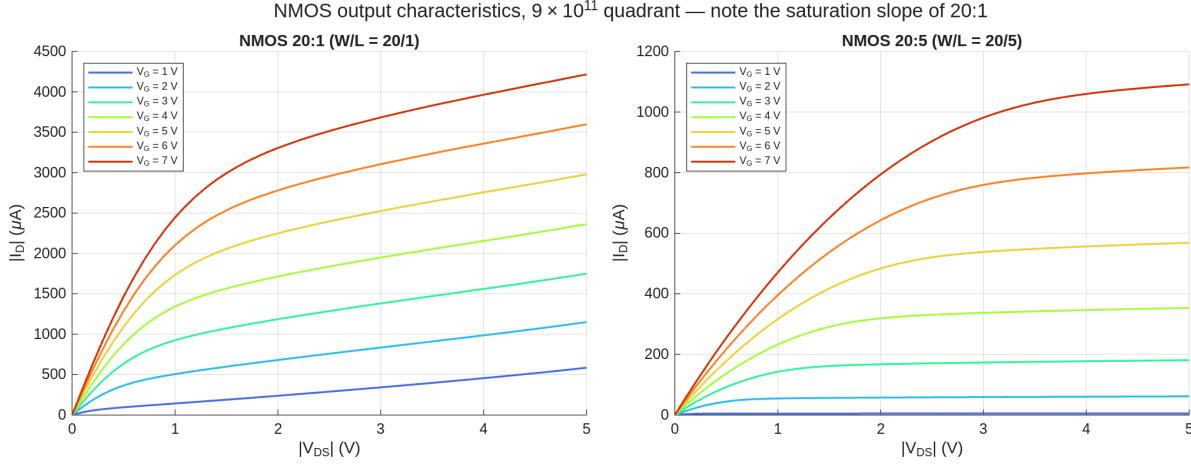


Figure 35: Measured NMOS output characteristics for the 20 : 1 and 20 : 5 devices (V_T -adjust dose $9 \times 10^{11} \text{ cm}^{-2}$).

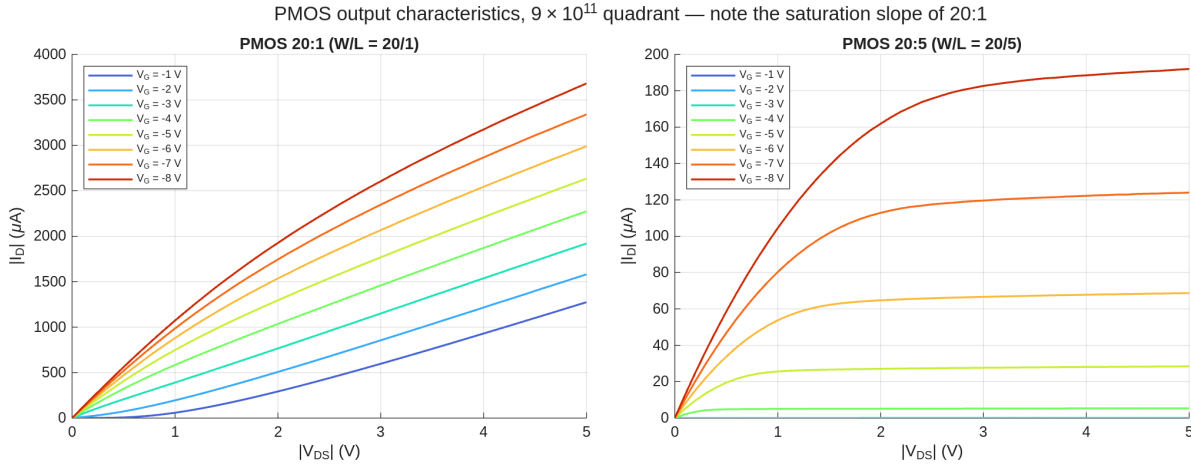


Figure 36: Measured PMOS output characteristics for the 20 : 1 and 20 : 5 devices (V_T -adjust dose $9 \times 10^{11} \text{ cm}^{-2}$).

Channel-length modulation parameters λ were extracted from linear fits of $|I_D|$ versus $|V_{\text{DS}}|$ in the saturation region, defined as $|V_{\text{DS}}| > |V_G - V_T| + 0.3 \text{ V}$. Representative values are shown in Table 15.

Besides the current magnitude, the 20:1 and 20:5 curves differ in shape, and two different short-channel effects are visible:

1. **Channel-length modulation and drain-field effects.** The pinch-off region length is set by the drain depletion width, so it consumes a fraction of the channel that scales with $1/L$. The long-channel devices show small λ values around 0.02 V^{-1} and essentially flat saturation, while for the 20:1 devices λ is roughly an order of magnitude larger. In the extreme case the drain field reaches the source: the PMOS 20:1 transistor never saturates within $|V_{\text{DS}}| \leq 5 \text{ V}$ and behaves almost like a resistor, consistent

Table 15: Channel-length modulation parameters extracted from the output characteristics.

Type	$W:L$	V_G (V)	λ (V^{-1})
NMOS	20:5	1 to 5	0.022–0.029
NMOS	20:1	2 to 4	0.16–0.42
PMOS	20:5	–4 to –7	0.015–0.018
PMOS	20:1	all	not extractable (no true saturation)

with DIBL and the onset of punch-through. Its deep ($0.59\mu\text{m}$) SP junctions in the lightly doped N-Well make the depletion regions as long as the drawn channel.

2. **Velocity saturation.** At $L = 1\mu\text{m}$ the lateral field reaches the range where the carrier drift velocity saturates. Its signatures are visible in Figure 37: the W/L -normalised current of the 20:1 device falls *below* that of the 20:5 device at the same gate voltage, and successive gate-voltage curves are spaced almost linearly instead of quadratically, i.e. $I_{D,\text{sat}}$ tends towards a linear dependence on $(V_{GS} - V_T)$ instead of the square law of Equation (15).

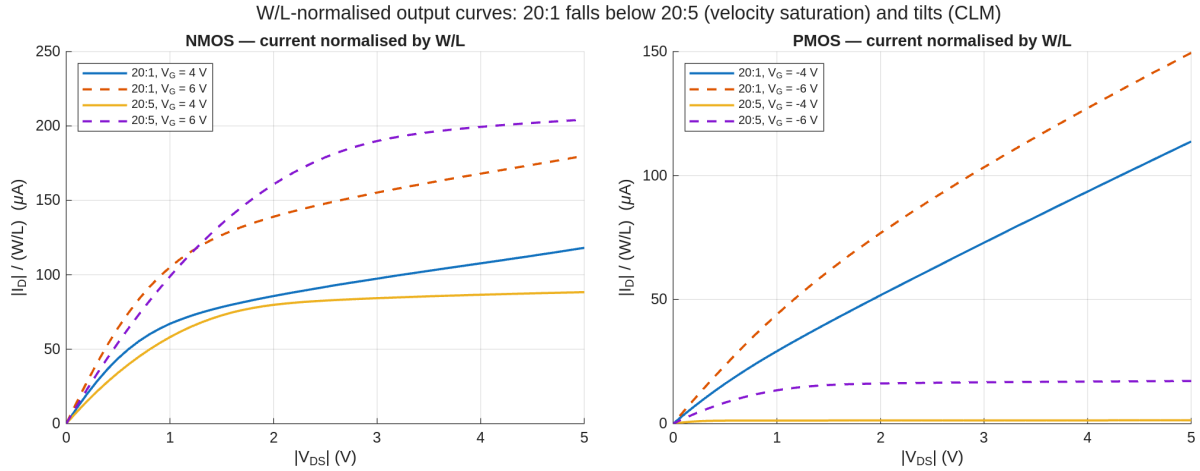


Figure 37: W/L -normalised output curves of the 20:1 and 20:5 devices. The short-channel device carries less normalised current (velocity saturation) and its saturation region tilts (channel-length modulation); the PMOS 20:1 device does not saturate at all.

These measurements reproduce the trends seen in the Sentaurus device simulations of Section 4: the simulated long-channel transistors follow the square law with a small saturation slope and only a mild velocity-saturation compression at the highest gate voltages, matching our clean 20:5 curves, while the strong short-channel degradation is only observed in the measured $L = 1\mu\text{m}$ devices, which have no simulated counterpart. Between the two measured device types, the PMOS short-channel device degrades far more severely than the NMOS, because of its deeper junctions and the roughly ten times lower doping of the N-Well compared to the NMOS channel region.

7 Conclusion

This report combined analytical calculations, process and device simulations, cleanroom processing, and electrical measurements to study the BICMOS5 process and the behaviour of the fabricated NMOS and PMOS devices. Overall, the results show good qualitative agreement between theory, Sentaurus simulations, and the experimental data, while the remaining differences can be explained by simplifying assumptions in the analytical models, the full thermal budget of the real process, and measurement-structure-dependent effects.

From the technology assignments and process simulations, the main trends of implantation, diffusion, and masking were confirmed. In crystalline silicon, channeling strongly affects implantation depth and profile

tails, whereas in amorphous silicon the dopant profiles become much closer to Gaussian. The simulations also showed that arsenic is the most suitable dopant for shallow, highly doped NMOS source/drain regions, while phosphorus is more suitable for the N-Well because it diffuses deeper into the substrate. In addition, the masking simulations showed that a thin ($0.4\text{ }\mu\text{m}$) photoresist layer is clearly insufficient for the chosen phosphorus implant and that an equally thick oxide layer only just stops it, while nitride provides the highest stopping power per unit thickness, although it is impractical as a standard implantation mask; in practice a sufficiently thick photoresist layer is used.

Comparison of the analytical junction-depth calculations with the Sentaurus process simulation showed clear differences. The simulated junction depths were approximately $0.537\text{ }\mu\text{m}$ for SN, $2.085\text{ }\mu\text{m}$ for the N-Well, and $0.594\text{ }\mu\text{m}$ for SP, whereas the simple calculations of Section 2 predicted much shallower SN and SP junctions (0.087 and $0.186\text{ }\mu\text{m}$) and a slightly deeper N-Well ($2.37\text{ }\mu\text{m}$). This difference is expected because the analytical treatment assumes simplified Gaussian or limited-source profiles and neglects the later thermal steps of the full CMOS process and the solid solubility limit, while the simulation includes the dopant redistribution during oxidation and annealing. Only for the N-Well, where the long drive-in dominates, is the analytical estimate close.

The processing assignments confirmed the expected behaviour of oxide growth, etching, and metal patterning. Wet oxidation produced a denser and more uniform oxide than PECVD, and the PECVD oxide etched about 3.2 times faster in buffered HF (5.53 versus 1.71 nm/s) because of its lower density and porosity. For the aluminium etch, wet etching showed very high selectivity to the underlying oxide but caused isotropic underetching, whereas dry etching gave straighter and better-defined metal lines at the cost of much lower oxide selectivity. These observations are consistent with the fundamental trade-off between selectivity and dimensional control in wet and dry processing.

The electrical measurements validated both the processing and the simulation results. Van der Pauw measurements gave sheet resistances of about $1740\text{ }\Omega/\square$ for the N-Well, $60.2\text{ }\Omega/\square$ for SN, and $488\text{ }\Omega/\square$ for SP, within 5 to 17% of the in-line four-point-probe values, and the aluminium interconnect measured $0.16\text{ }\Omega/\square$, corresponding to a resistivity of $3.2 \times 10^{-6}\text{ }\Omega\text{cm}$, the literature value for Al with 1% Si. The average active dopant concentrations extracted through the Irvin curves (1.5×10^{16} , 2.3×10^{19} and $1.6 \times 10^{18}\text{ cm}^{-3}$ for N-Well, SN and SP) agree with the Section 5 values to within about 10%, and are only consistent with the simulated junction depths, not the analytical ones. The ELM measurements showed clear lateral out-diffusion, which was small for SN, about $0.15\text{ }\mu\text{m}$ per side, but much larger for the N-Well, on the order of 0.8 to $2.1\text{ }\mu\text{m}$ per side, consistent with the longer and hotter phosphorus drive-in. This confirms that lateral diffusion becomes increasingly important for lightly doped, deeply driven layers.

The transistor measurements also matched the expected device physics. For long-channel devices, the extracted threshold voltages and mobilities remained nearly constant with geometry, confirming that they are process rather than device parameters, with electron mobilities around $750\text{--}800\text{ cm}^2/\text{Vs}$ and hole mobilities around $230\text{--}300\text{ cm}^2/\text{Vs}$, both below bulk values as expected for surface-channel conduction. The short-channel ($L = 1\text{ }\mu\text{m}$) devices showed threshold-voltage roll-off, an artificially inflated apparent mobility, roughly ten times stronger channel-length modulation, velocity-saturation signatures in the normalised output curves, and in the PMOS case the onset of punch-through, all in agreement with the trends predicted by device theory and the simulations.

Finally, the influence of the V_T -adjust implant was clearly demonstrated. Increasing the boron dose from 0 to $9 \times 10^{11}\text{ cm}^{-2}$ increased the measured NMOS threshold voltage monotonically from about -0.034 V to 0.819 V , while the PMOS threshold voltage became less negative, changing from about -3.702 V to -3.300 V . The Sentaurus device simulations reproduce both trends and the dose sensitivity, with a systematic offset of about 0.5 to 0.6 V that is attributed to the different threshold-extraction criteria and to the oxide and interface charge that is not included in the simulation. This confirms the theoretical expectation that a single unmasked boron implant raises the channel acceptor concentration for the NMOS and partially compensates the N-Well surface doping for the PMOS. Therefore, the measurements, simulations, and theoretical analysis consistently show how implantation and thermal processing determine the final electrical behaviour of the BICMOS5 devices.

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